

Initiating Search

October 21, 2025, 2:36 PM

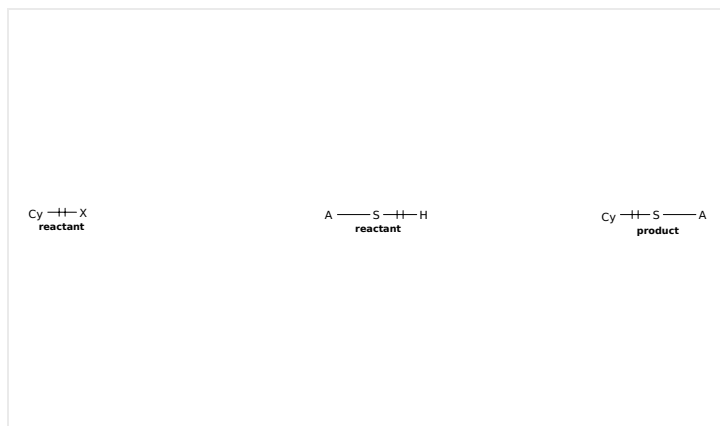
• Search:

Filtered By:

Yield: 90-100%, 80-89%, 70-79%

Document Type: Journal

Publication Year: 2015 to 2025



Structure Match: Substructure

Search Tasks

Task	Result Type	View
Exported: Returned Reaction Results + Filters (6,540)	 Reactions	View Results

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

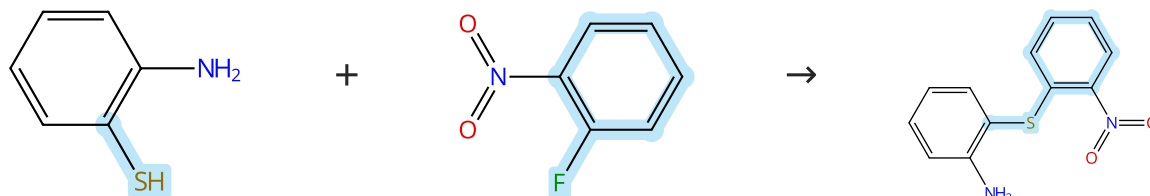
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

Reactions (100)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (67)

Suppliers (93)

Suppliers (51)

31-614-CAS-43952833

Steps: 1 Yield: 100%

Synthesis and Reactivity of Six-Membered Cyclic Diaryl λ^3 -Bromanes and λ^3 -Chloranes

By: Bhattacharjee, Dhananjay; et al

Angewandte Chemie, International Edition (2025), 64(14), e202424559.

1.1 Reagents: Potassium hydroxide

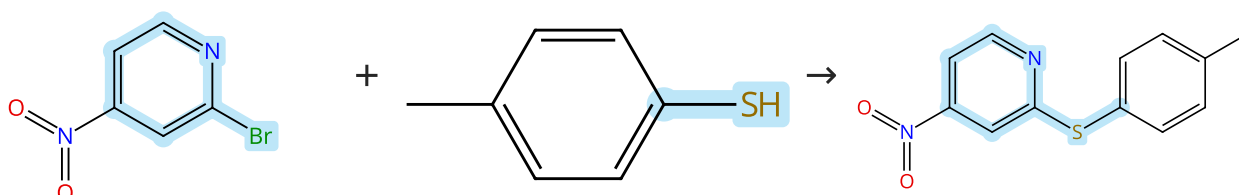
Solvents: Acetonitrile; 15 min, rt

1.2 overnight, 110 °C

Experimental Protocols

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (56)

31-171-CAS-22372702

Steps: 1 Yield: 100%

C-S coupling with nitro group as leaving group via simple inorganic salt catalysis

By: Xuan, Maojie; et al

Chinese Chemical Letters (2020), 31(1), 84-90.

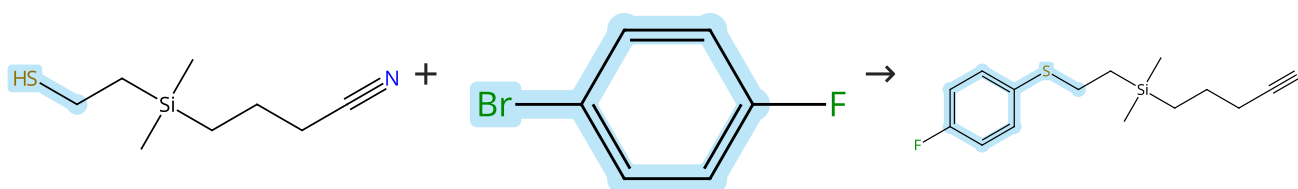
1.1 Catalysts: Tripotassium phosphate, 18-Crown-6

Solvents: Tetrahydrofuran; 3 h, rt

Experimental Protocols

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

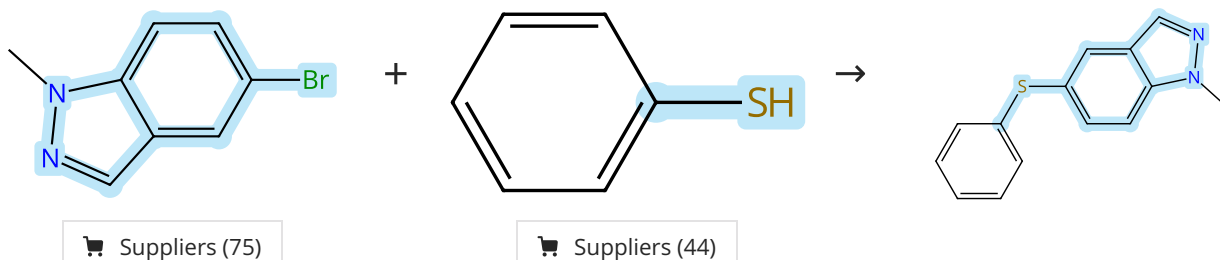


Suppliers (103)

31-171-CAS-20938538 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: 1,4-Dioxane; 5 min, rt 1.2 Catalysts: Dichlorobis(triphenylphosphine)palladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]; 12 h, 110 °C	2-(3-Cyanopropyl)dimethylsilyl)ethyl as a Polar Sulfur Protecting Group By: Bannwart, Linda M.; et al Synthesis (2019), 51(22), 4153-4164.
---	--

Scheme 4 (1 Reaction)

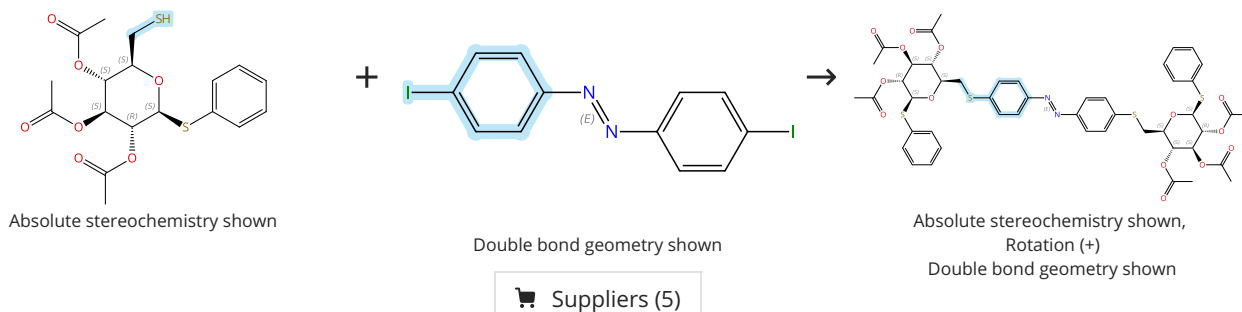
Steps: 1 Yield: 100%



31-171-CAS-20461552 Steps: 1 Yield: 100% 1.1 Reagents: Lithium bis(trimethylsilyl)amide Catalysts: Palladium(1+), [2'-(amino-κM)[1,1'-biphenyl]-2-yl-κC] [[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl-κC']bis(1,1-dimethylethyl)phosphine-κP]-, (<i>SP-4-2</i>)-, 7,7-dimethyl-2-oxobicyclo[2.2.1]heptane-1-methanesulfonate (1: 1) Solvents: Tetrahydrofuran; 2 h, rt	Monophosphine Ligands Promote Pd-Catalyzed C-S Cross-Coupling Reactions at Room Temperature with Soluble Bases By: Xu, Jessica; et al ACS Catalysis (2019), 9(7), 6461-6466.
--	--

Scheme 5 (1 Reaction)

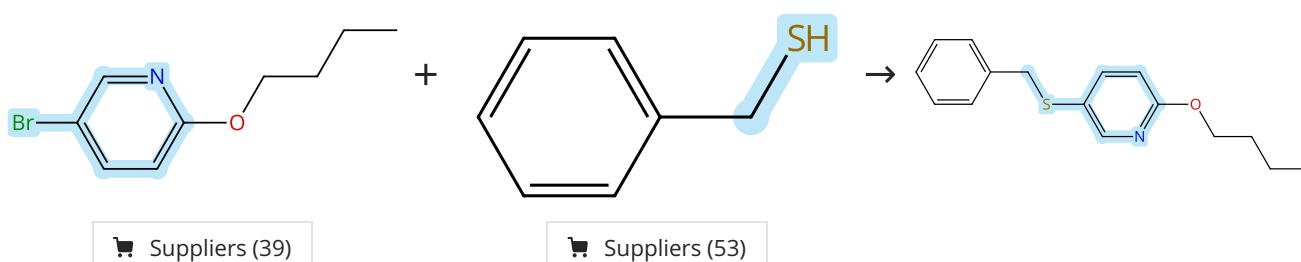
Steps: 1 Yield: 100%



31-614-CAS-36707287 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[5-(diphenylphosphino)-9,9-dimethyl-9<i>H</i>-xanthene-4-yl]diphenylphosphine-κP](methanesulfonato-κO)palladium Solvents: Tetrahydrofuran; 10 min, rt Experimental Protocols	Sulfur and azobenzenes, a profitable liaison: straight forward synthesis of photo-switchable thioglycosides with tunable properties By: Berry, Jonathan; et al Chemistry - A European Journal (2022), 28(39), e202200354.
--	---

Scheme 6 (1 Reaction)

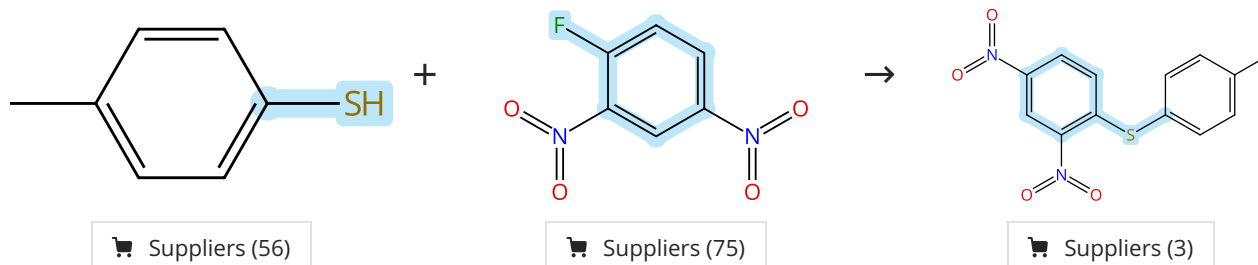
Steps: 1 Yield: 100%



31-614-CAS-37046538 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; 4 h, reflux Experimental Protocols	Catalytic Asymmetric Hydrogen Atom Transfer: Enantioselective Hydroamination of Alkenes By: Hejna, Benjamin G.; et al Journal of the American Chemical Society (2023), 145(29), 16118-16129.
---	--

Scheme 7 (1 Reaction)

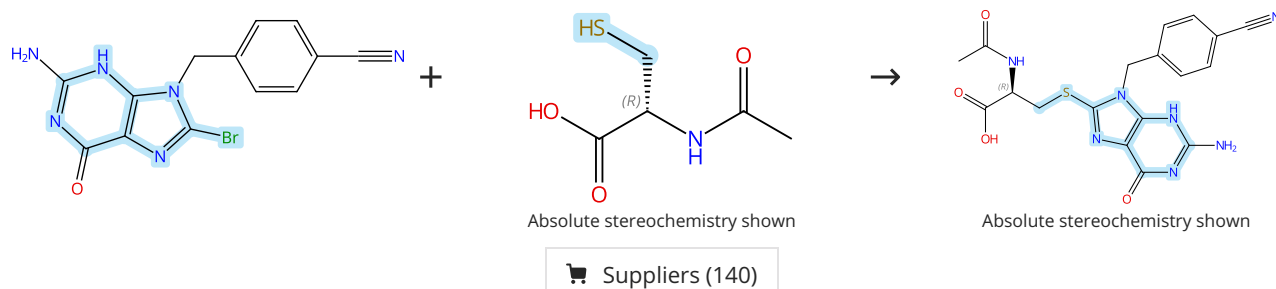
Steps: 1 Yield: 100%



31-171-CAS-22372699 Steps: 1 Yield: 100% 1.1 Catalysts: Tripotassium phosphate, 18-Crown-6 Solvents: Tetrahydrofuran; 3 h, rt Experimental Protocols	C-S coupling with nitro group as leaving group via simple inorganic salt catalysis By: Xuan, Maojie; et al Chinese Chemical Letters (2020), 31(1), 84-90.
---	---

Scheme 8 (1 Reaction)

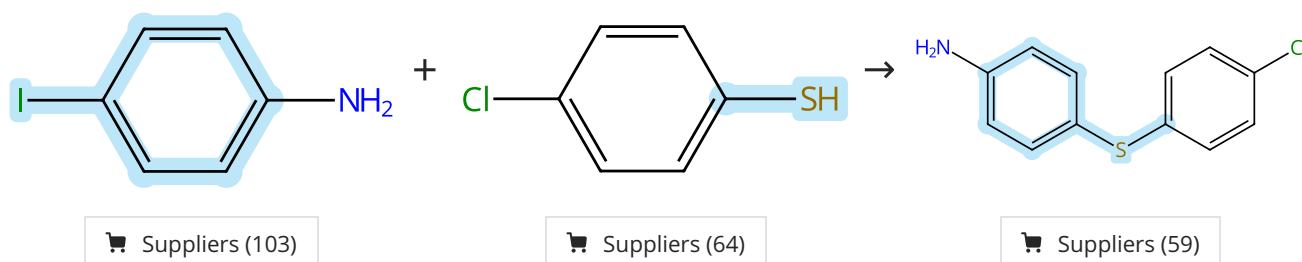
Steps: 1 Yield: 100%



31-614-CAS-37526770 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; overnight, 80 °C 1.2 Reagents: Hydrochloric acid Solvents: Water; acidified; 2 h, rt Experimental Protocols	Second-Generation AUTACs for Targeted Autophagic Degradation By: Takahashi, Daiki; et al Journal of Medicinal Chemistry (2023), 66(17), 12342-12372.
---	--

Scheme 9 (1 Reaction)

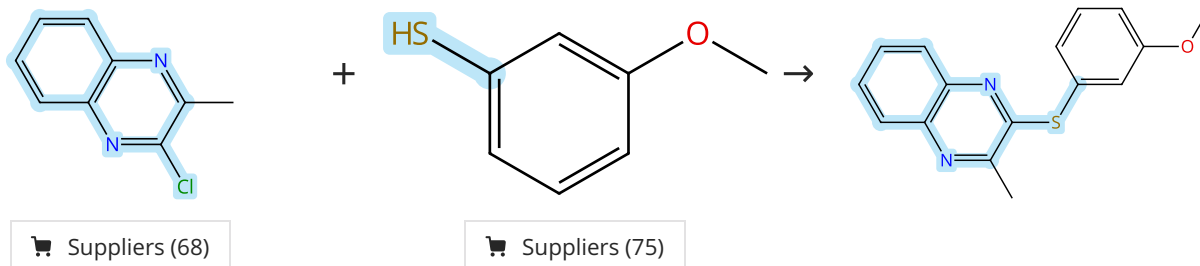
Steps: 1 Yield: 100%



31-614-CAS-31632129	Steps: 1 Yield: 100%	Rationale, synthesis and biological evaluation of substituted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site By: Gagne-Boulet, Mathieu; et al Journal of Molecular Structure (2022), 1259, 132691.
1.1 Reagents: Cesium carbonate Catalysts: Copper oxide (CuO) Solvents: Toluene; 24 h, 110 °C		
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-31987819	Steps: 1 Yield: 100%	Antiplasmodial 2-thiophenoxy-3-trichloromethyl quinoxalines target the apicoplast of Plasmodium falciparum By: Amrane, Dyhia; et al European Journal of Medicinal Chemistry (2021), 224, 113722.
1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; overnight, 70 °C		
Experimental Protocols		

Scheme 11 (1 Reaction)

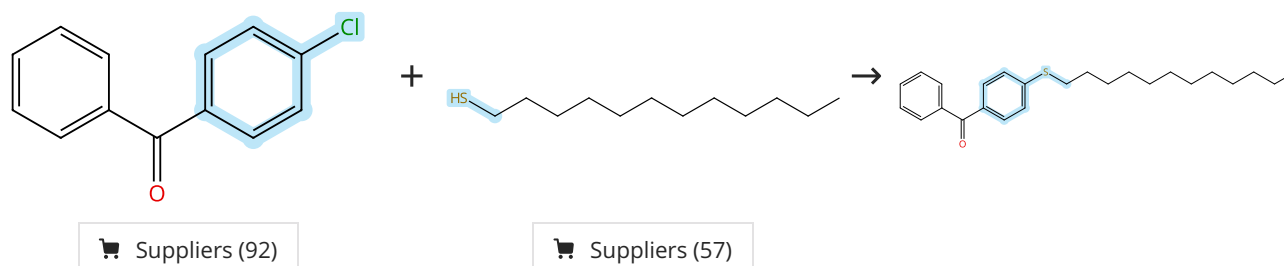
Steps: 1 Yield: 100%



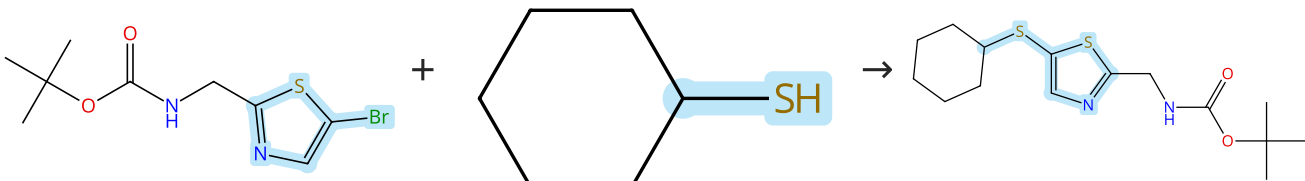
31-614-CAS-40567095	Steps: 1 Yield: 100%	Highly selective scalable electrosynthesis of 4-hydroxybenzo[e]-1,2,4-thiadiazine-1,1-dioxides By: Winter, Johannes; et al Cell Reports Physical Science (2024), 5(5), 101927.
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 15 min, rt		
1.2 -		
Experimental Protocols		

Scheme 12 (1 Reaction)

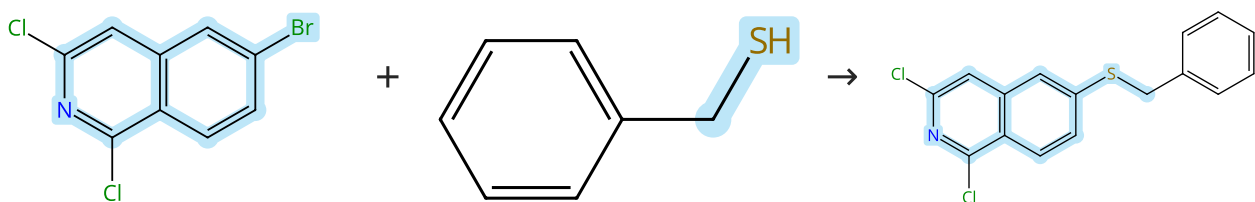
Steps: 1 Yield: 100%



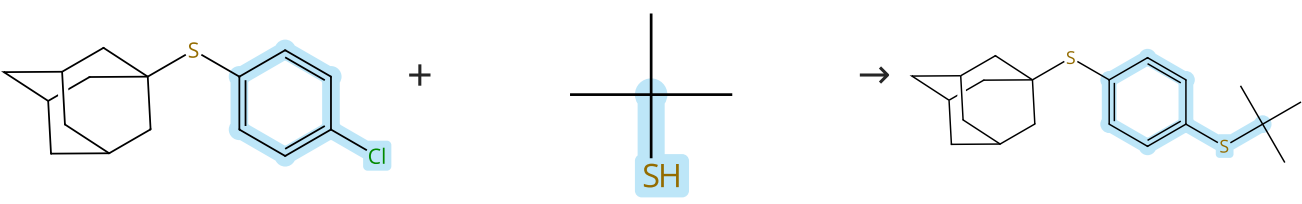
31-614-CAS-37149592 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 6 h, 0 °C 1.2 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Syntheses of aryl thioethers via aromatic substitution of aryl halides at 0 to 25°C By: Tsuzaki, Marina; et al Chemical & Pharmaceutical Bulletin (2023), 71(7), 620-623.
--	---

Scheme 13 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (48)  Suppliers (35)  Supplier (1)		
---	--	--

31-171-CAS-21642215 Steps: 1 Yield: 100% 1.1 Reagents: Sodium <i>tert</i>-butoxide Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: <i>tert</i>-Butanol, Toluene; 18 h, 110 °C	2-Aminomethylene-5-sulfonylthiazole Inhibitors of Lysyl Oxidase (LOX) and LOXL2 Show Significant Efficacy in Delaying Tumor Growth By: Smithen, Deborah A.; et al Journal of Medicinal Chemistry (2020), 63(5), 2308-2324.
---	--

Scheme 14 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (61)  Suppliers (53)  Suppliers (3)		
--	--	--

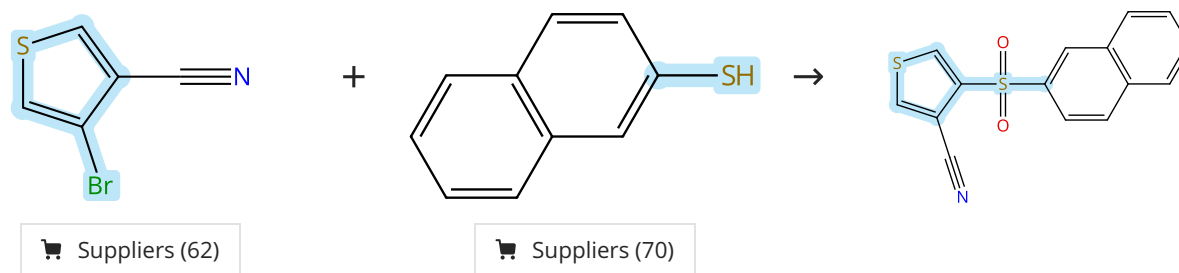
31-171-CAS-17394836 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; 25 min, 60 °C; 60 °C → rt Experimental Protocols	Sulfonamides as Selective Nav1.7 Inhibitors: Optimizing Potency and Pharmacokinetics While Mitigating Metabolic Liabilities By: Weiss, Matthew M.; et al Journal of Medicinal Chemistry (2017), 60(14), 5969-5989.
---	--

Scheme 15 (1 Reaction) Steps: 1 Yield: 100%   Suppliers (50)		
---	--	--

31-614-CAS-31329038 Steps: 1 Yield: 100% 1.1 Reagents: Potassium acetate Catalysts: (<i>SP</i>-5-34)-Chloro[1,1'-(9,9-dimethyl-9<i>H</i>-xanthene-4,5-diyl-κO)bis[1,1-diphenylphosphine-κP](2-methylphenyl)nickel Solvents: Tetrahydrofuran; 2 h, rt 1.2 Solvents: Water; rt Experimental Protocols	Acetate Facilitated Nickel Catalyzed Coupling of Aryl Chlorides and Alkyl Thiols By: Oechsner, Regina M.; et al ACS Catalysis (2022), 12(4), 2233-2243.
---	--

Scheme 16 (1 Reaction)

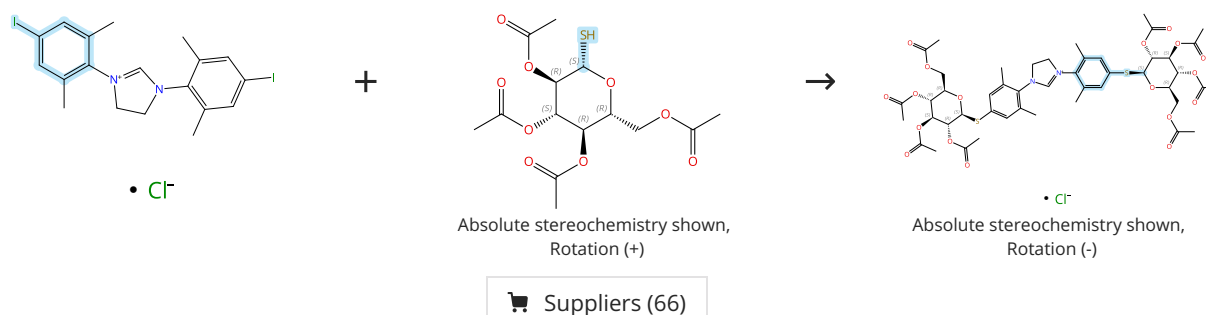
Steps: 1 Yield: 100%



31-144-CAS-20264624 Steps: 1 Yield: 100% 1.1 Reagents: Sodium <i>tert</i>-butoxide Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: Toluene; 16 h, 110 °C; 110 °C → rt 1.2 Reagents: <i>m</i>-Chloroperbenzoic acid Solvents: Dichloromethane; 0 °C; 3 h, rt	Anti-metastatic Inhibitors of Lysyl Oxidase (LOX): Design and Structure-Activity Relationships By: Leung, Leo; et al Journal of Medicinal Chemistry (2019), 62(12), 5863-5884.
--	---

Scheme 17 (1 Reaction)

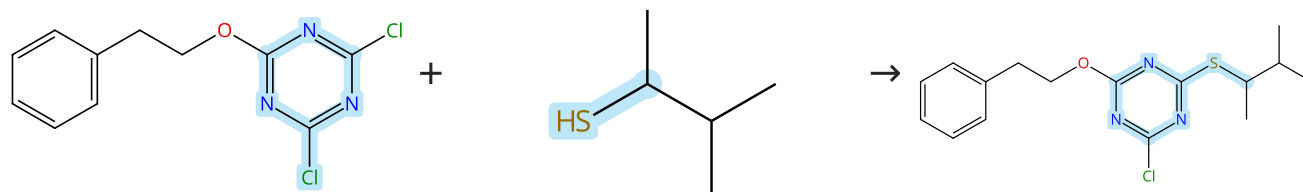
Steps: 1 Yield: 100%



31-614-CAS-34373388 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine Catalysts: [2'-(Amino-κM)[1,1'-biphenyl]-2-yl-κC][[5-(diphenylphosphino)-9,9-dimethyl-9<i>H</i>-xanthen-4-yl]diphenylphosphine-κP](methanesulfonato-κO)palladium Solvents: Tetrahydrofuran; 5 min, rt 1.2 Solvents: Tetrahydrofuran; 20 min, rt; 30 min, rt Experimental Protocols	Azoliums and Ag(I)-N-Heterocyclic Carbene Thioglycosides: Synthesis, Reactivity and Bioactivity By: Ryzhakov, Dmytro; et al European Journal of Organic Chemistry (2022), 2022(9), e202101499.
---	---

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (4)

Suppliers (47)

31-171-CAS-20778206

Steps: 1 Yield: 100%

Exploring the orthogonal chemoselectivity of 2,4,6-trichloro-1,3,5-triazine (TCT) as a trifunctional linker with different nucleophiles: rules of the game

1.1 Reagents: Diisopropylethylamine
Solvents: Dichloromethane; 12 h, rt

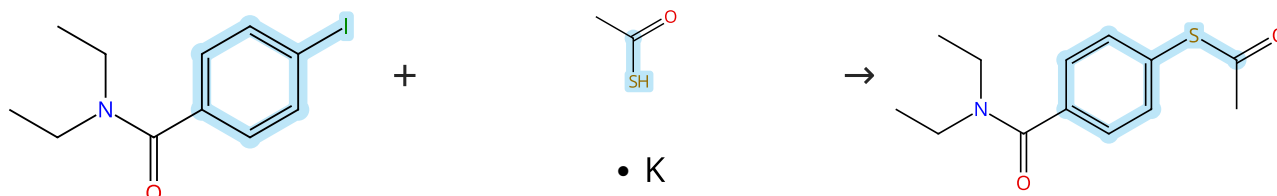
Experimental Protocols

By: Sharma, Anamika; et al

Frontiers in Chemistry (Lausanne, Switzerland) (2018), 6, 516pp..

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (45)

Suppliers (84)

31-614-CAS-40340401

Steps: 1 Yield: 100%

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C

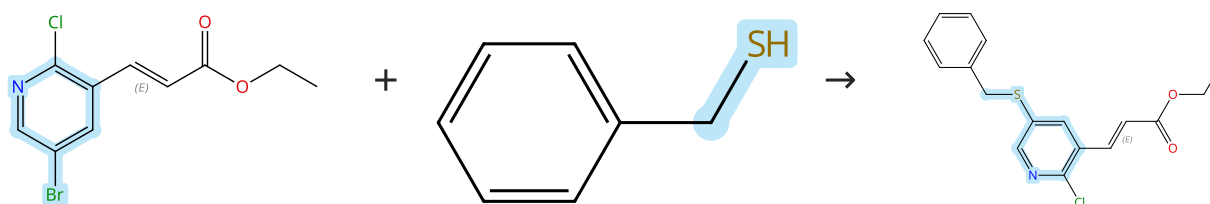
Experimental Protocols

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Double bond geometry shown

Supplier (1)

Suppliers (53)

Double bond geometry shown

Supplier (1)

31-171-CAS-17458280

Steps: 1 Yield: 100%

Sulfonamides as Selective Nav1.7 Inhibitors: Optimizing Potency, Pharmacokinetics, and Metabolic Properties to Obtain Atropisomeric Quinolinone (AM-0466) that Affords Robust in Vivo Activity

1.1 Reagents: Diisopropylethylamine
Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]
Solvents: 1,4-Dioxane; 1 h, 80 °C

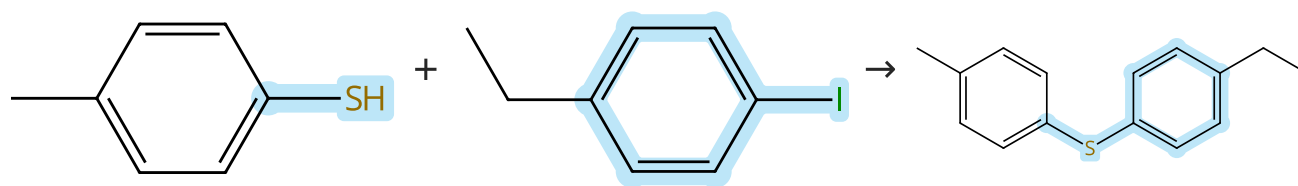
Experimental Protocols

By: Graceffa, Russell F.; et al

Journal of Medicinal Chemistry (2017), 60(14), 5990-6017.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (56)

Suppliers (71)

Suppliers (5)

31-171-CAS-23873774

Steps: 1 Yield: 100%

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

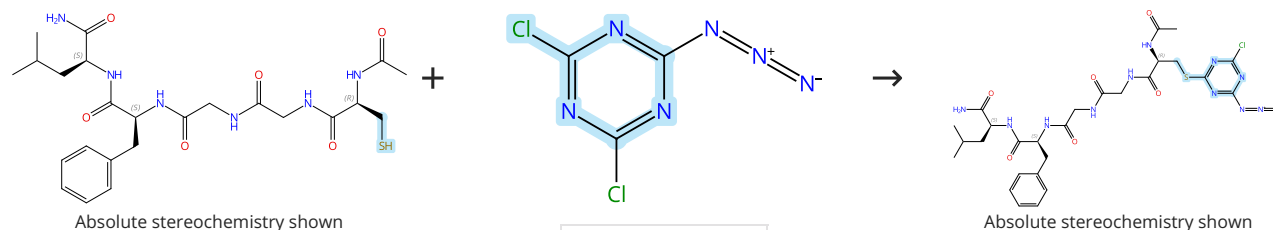
Journal of Chemical Crystallography (2021), 51(3), 301-310.

- 1.1 **Reagents:** Potassium hydroxide
Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C
- 1.2 **Catalysts:** Cuprous iodide; 12 h, 110 °C

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (4)

31-614-CAS-34441023

Steps: 1 Yield: 100%

Exploiting azido-dichloro-triazine as a linker for regioselective incorporation of peptides through their N, O, S functional groups

By: Sharma, Anamika; et al

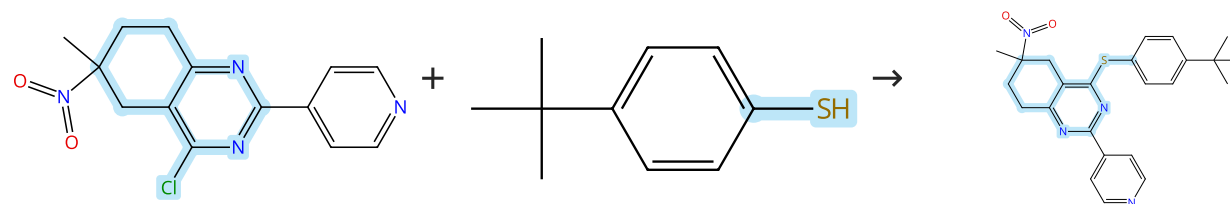
Bioorganic Chemistry (2020), 104, 104334.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Acetonitrile, Water; 30 min, > pH 8, 0 °C

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (74)

31-614-CAS-39352334

Steps: 1 Yield: 100%

Investigation into a Tetrahydroquinazoline Scaffold

By: Jardner, Jonathan; et al

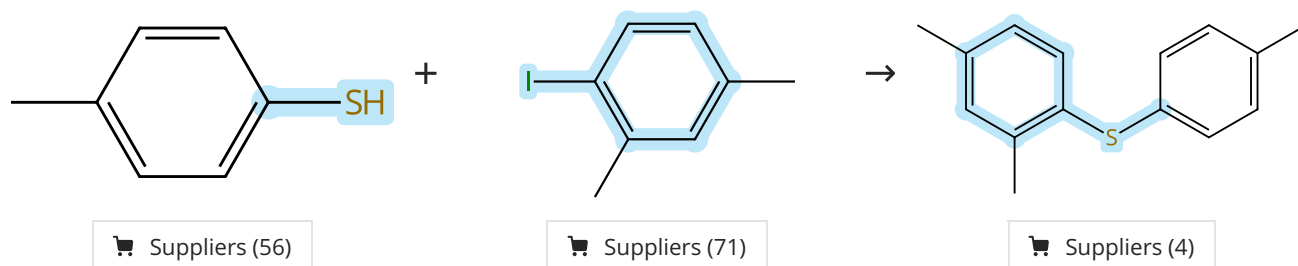
European Journal of Organic Chemistry (2024), 27(10), e202400050.

- 1.1 **Reagents:** Triethylamine
Solvents: 1,4-Dioxane; 18 h, 100 °C

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



31-171-CAS-23874822

Steps: 1 Yield: 100%

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

1.1 Reagents: Potassium hydroxide

Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C

By: Zhou, Wen-Yan; et al

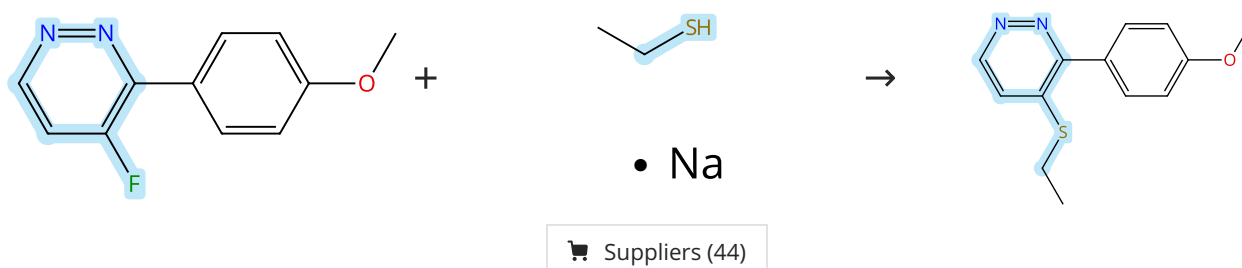
1.2 Catalysts: Cuprous iodide; 12 h, 110 °C

Journal of Chemical Crystallography (2021), 51(3), 301-310.

Experimental Protocols

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



31-171-CAS-17683936

Steps: 1 Yield: 100%

Synthesis of Functionalized 4-Fluoropyridazines

1.1 Solvents: Acetonitrile; 24 h, 50 °C

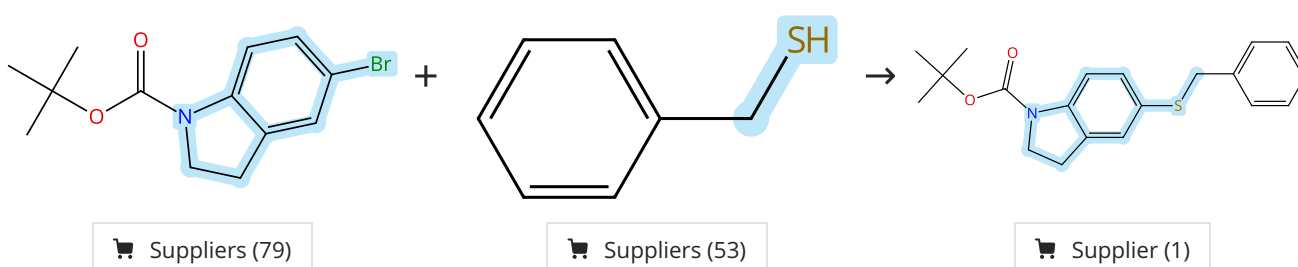
By: Feraldi-Xypolia, Alexandra; et al

Experimental Protocols

Asian Journal of Organic Chemistry (2017), 6(7), 927-935.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



31-171-CAS-4146912

Steps: 1 Yield: 100%

Discovery of a Novel Series of N-Phenylindoline-5-sulfonamide Derivatives as Potent, Selective, and Orally Bioavailable Acyl CoA:Monoacylglycerol Acyltransferase-2 Inhibitors

1.1 Reagents: Diisopropylethylamine

Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]

Solvents: Toluene; 5 h, reflux

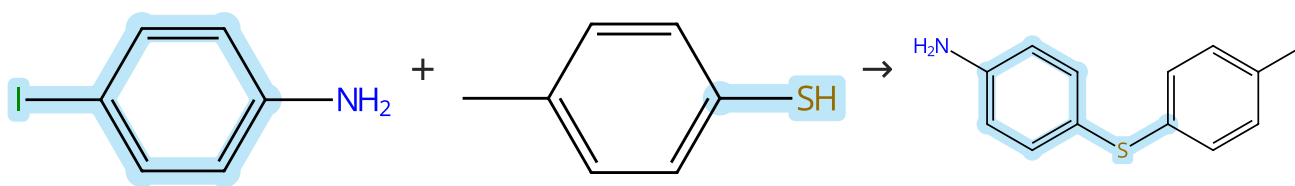
By: Sato, Kenjiro; et al

Journal of Medicinal Chemistry (2015), 58(9), 3892-3909.

Experimental Protocols

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (103)

Suppliers (56)

Suppliers (45)

31-614-CAS-31632127

Steps: 1 Yield: 100%

1.1 Reagents: Cesium carbonate
Catalysts: Copper oxide (CuO)
Solvents: Toluene; 24 h, 110 °C

Experimental Protocols

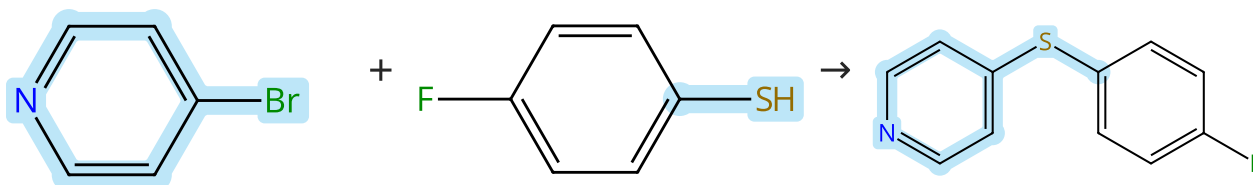
Rationale, synthesis and biological evaluation of substituted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site

By: Gagne-Boulet, Mathieu; et al

Journal of Molecular Structure (2022), 1259, 132691.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (40)

Suppliers (78)

Suppliers (6)

31-171-CAS-20519064

Steps: 1 Yield: 100%

1.1 Reagents: Potassium carbonate
Solvents: Dimethyl sulfoxide; overnight, 110 °C

1.2 Reagents: Ammonium chloride
Solvents: Water

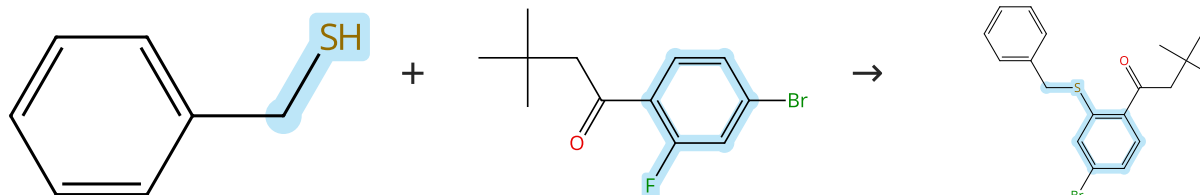
Structure-Based Discovery of M-89 as a Highly Potent Inhibitor of the Menin-Mixed Lineage Leukemia (Menin-MLL) Protein-Protein Interaction

By: Aguilar, Angelo; et al

Journal of Medicinal Chemistry (2019), 62(13), 6015-6034.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (53)

Suppliers (30)

Supplier (1)

31-171-CAS-19716776

Steps: 1 Yield: 100%

1.1 Reagents: Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; rt; 5 min, rt

1.2 Solvents: Tetrahydrofuran; 30 min, rt

Experimental Protocols

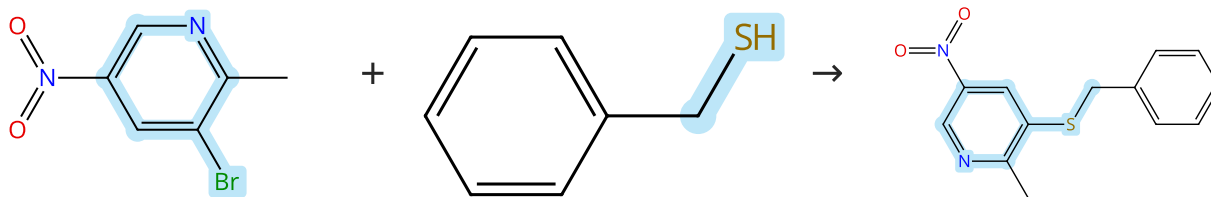
Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

Suppliers (53)

31-614-CAS-33617322

Steps: 1 Yield: 100%

Discovery of a First-in-Class Inhibitor of the PRMT5-Substrate Adaptor Interaction

By: McKinney, David C.; et al

Journal of Medicinal Chemistry (2021), 64(15), 11148-11168.

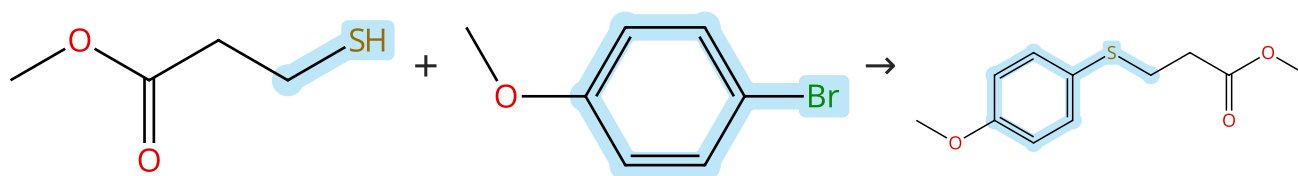
1.1 Reagents: Diisopropylethylamine

Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]

Solvents: 1,4-Dioxane; 12 h, 100 °C

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (71)

Suppliers (6)

31-614-CAS-34820788

Steps: 1 Yield: 100%

Modulating the Surface and Photophysical Properties of Carbon Dots to Access Colloidal Photocatalysts for Cross-Couplings

By: Zhao, Zhouxiang; et al

ACS Catalysis (2022), 12(22), 13831-13837.

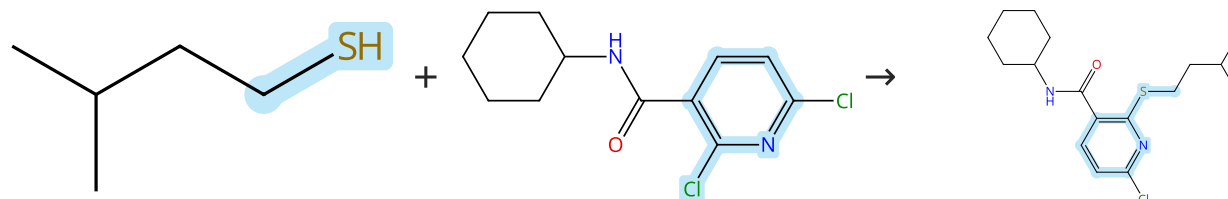
1.1 Catalysts: Nickel bromide, Pyridine, hydriodide (1:1), 4,4'-Bis(1,1-dimethylethyl)-2,2'-bipyridine

Solvents: Dimethylacetamide; 10 min, rt; 90 h, rt

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (40)

Suppliers (22)

Supplier (1)

31-171-CAS-17114910

Steps: 1 Yield: 100%

Discovery and Optimization of Allosteric Inhibitors of Mutant Isocitrate Dehydrogenase 1 (R132H IDH1) Displaying Activity in Human Acute Myeloid Leukemia Cells

By: Jones, Stuart; et al

Journal of Medicinal Chemistry (2016), 59(24), 11120-11137.

1.1 Reagents: Sodium bicarbonate

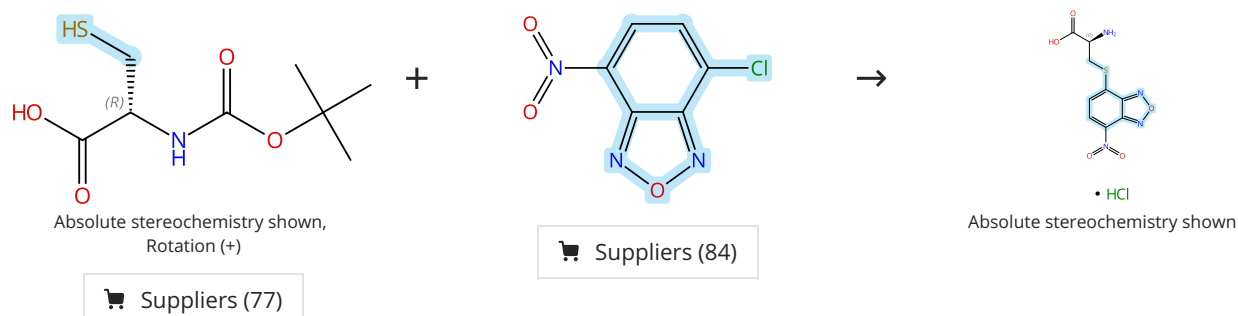
Solvents: Dimethylformamide; 45 min, rt

1.2 Solvents: Dimethylformamide; 0 °C; 4 h, 60 °C

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



31-614-CAS-39061034

Steps: 1 Yield: 100%

Small Fluorogenic Amino Acids for Peptide-Guided Background-Free Imaging

By: de Moliner, Fabio; et al

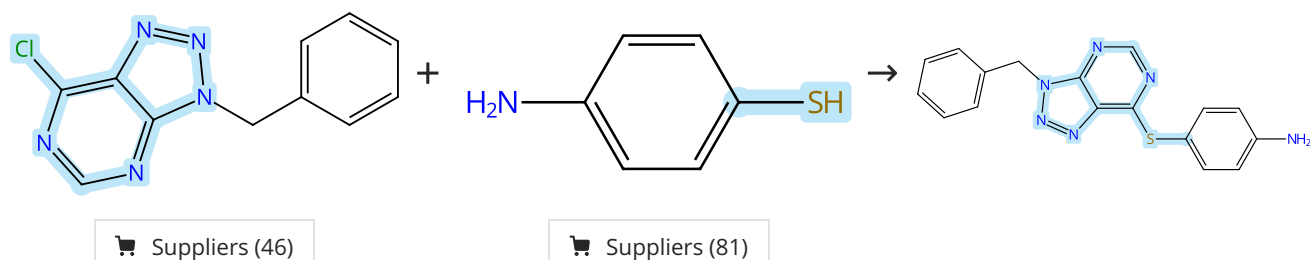
Angewandte Chemie, International Edition (2023), 62(4), e202216231.

- 1.1 Reagents: Sodium bicarbonate
Solvents: Methanol, Water; 55 °C; 15 min, 55 °C
- 1.2 Reagents: Hydrochloric acid
Solvents: 1,4-Dioxane; 30 min, rt

Experimental Protocols

Scheme 34 (2 Reactions)

Steps: 1 Yield: 100%



31-614-CAS-44577063

Steps: 1 Yield: 100%

Design of Benzyl-triazolopyrimidine-Based NADPH Oxidase Inhibitors Leads to the Discovery of a Potent Dual Covalent N OX2/MAOB Inhibitor

By: Noce, Beatrice; et al

Journal of Medicinal Chemistry (2025), 68(6), 6292-6311.

- 1.1 Reagents: Triethylamine
Solvents: Ethanol; 1 h, rt
- 1.2 Reagents: Sodium chloride

Experimental Protocols

31-614-CAS-44576984

Steps: 1 Yield: 100%

Design of Benzyl-triazolopyrimidine-Based NADPH Oxidase Inhibitors Leads to the Discovery of a Potent Dual Covalent N OX2/MAOB Inhibitor

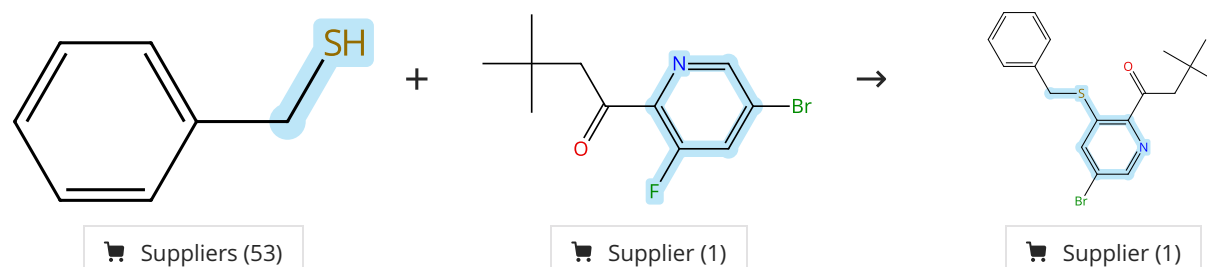
By: Noce, Beatrice; et al

Journal of Medicinal Chemistry (2025), 68(6), 6292-6311.

- 1.1 Reagents: Triethylamine
Solvents: Ethanol; 0 °C; 3 h, rt
- 1.2 Reagents: Diethyl ether

Scheme 35 (1 Reaction)

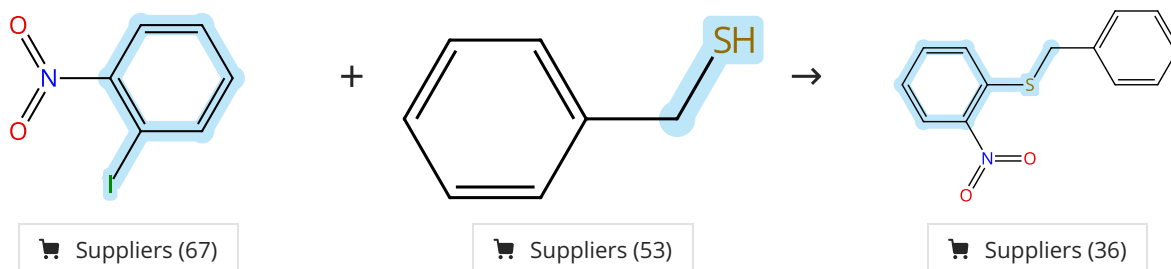
Steps: 1 Yield: 100%



31-171-CAS-19716820 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Tetrahydrofuran; rt; 5 min, rt 1.2 Solvents: Tetrahydrofuran; 30 min, rt Experimental Protocols	Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄) By: Bollinger, Sean R.; et al Journal of Medicinal Chemistry (2019), 62(1), 342-358.
--	---

Scheme 36 (1 Reaction)

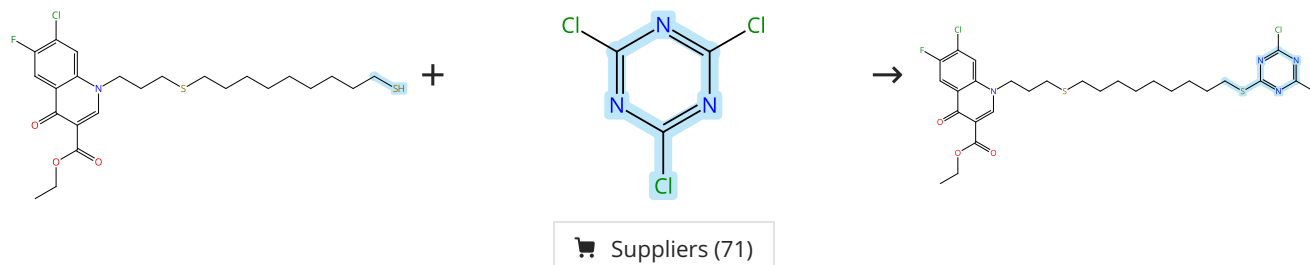
Steps: 1 Yield: 100%



31-171-CAS-18817232 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Catalysts: Bis(L-prolinato-κN,κO)palladium Solvents: Ethanol; 6 h, 70 °C	C-S Cross-Coupling Reaction Using a Recyclable Palladium Prolinate Catalyst under Mild and Green Conditions By: Santos, Beatriz F.; et al ChemistrySelect (2017), 2(28), 9063-9068.
--	--

Scheme 37 (1 Reaction)

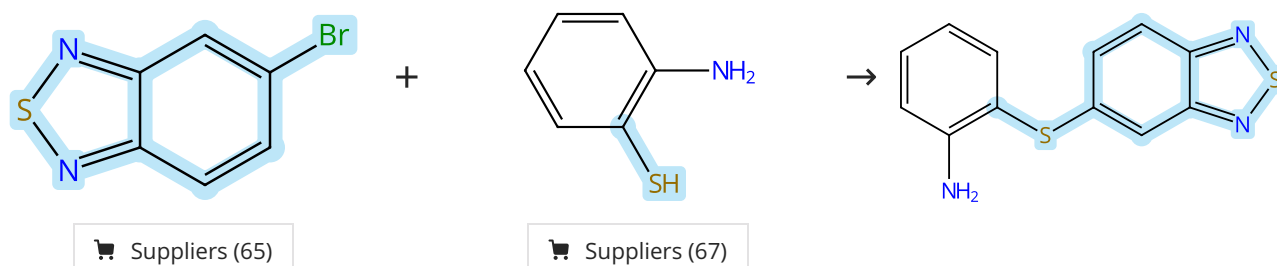
Steps: 1 Yield: 100%



31-614-CAS-25078017 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Tetrahydrofuran; 0 °C	Triazine mediated covalent antibiotic grafting on cotton fabrics as a modular approach for developing antimicrobial barriers By: Montagut, Ana Maria; et al Cellulose (Dordrecht, Netherlands) (2019), 26(12), 7495-7505.
---	--

Scheme 38 (1 Reaction)

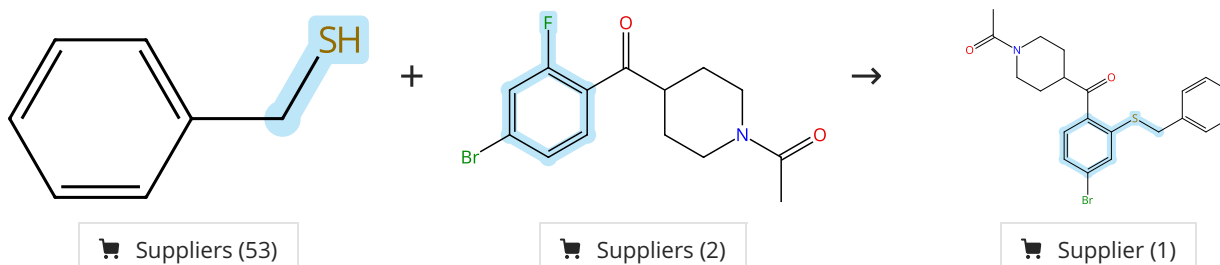
Steps: 1 Yield: 100%



31-171-CAS-21928138 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Catalysts: Bis(L-prolinato-κN,κO)palladium Solvents: Dimethylformamide; 6 h, 100 °C Experimental Protocols	Efficient palladium-catalyzed C-S cross-coupling reaction of benzo-2,1,3-thiadiazole at C-5-position: A potential class of A ChE inhibitors By: dos Santos, Beatriz F.; et al Applied Organometallic Chemistry (2020), 34(7), e5650.
---	---

Scheme 39 (1 Reaction)

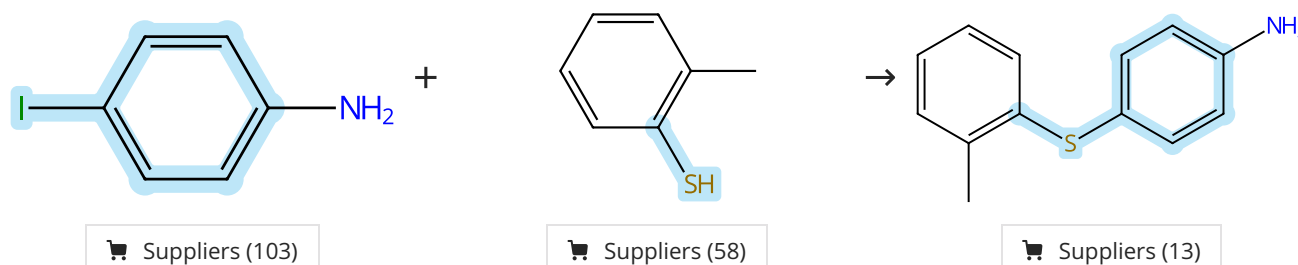
Steps: 1 Yield: 100%



31-171-CAS-19716796 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Tetrahydrofuran; rt; 5 min, rt 1.2 Solvents: Tetrahydrofuran; 30 min, rt Experimental Protocols	Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄) By: Bollinger, Sean R.; et al Journal of Medicinal Chemistry (2019), 62(1), 342-358.
--	---

Scheme 40 (1 Reaction)

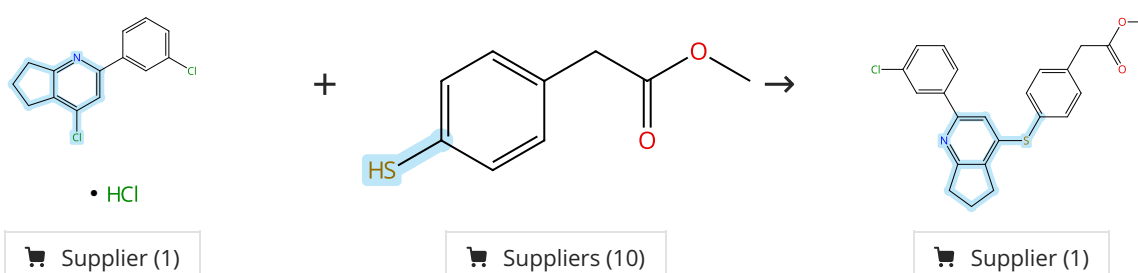
Steps: 1 Yield: 100%



31-614-CAS-31632124 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Copper oxide (CuO) Solvents: Toluene; 24 h, 110 °C Experimental Protocols	Rationale, synthesis and biological evaluation of substituted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site By: Gagne-Boulet, Mathieu; et al Journal of Molecular Structure (2022), 1259, 132691.
--	--

Scheme 41 (1 Reaction)

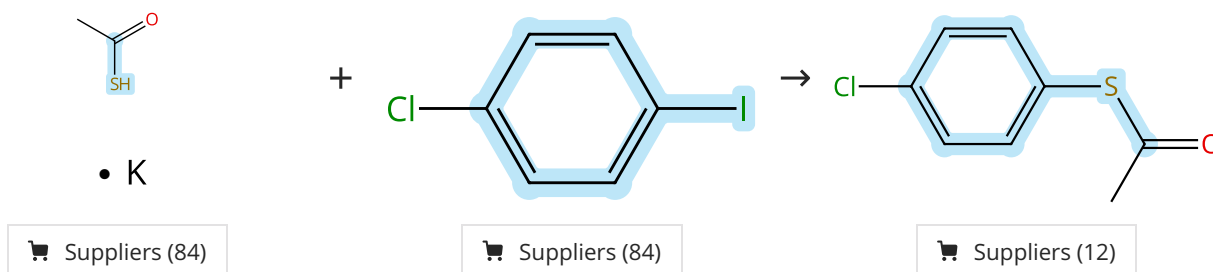
Steps: 1 Yield: 100%



31-171-CAS-20432413	Steps: 1 Yield: 100%	Design and Synthesis of Selective Phosphodiesterase 4D (PD E4D) Allosteric Inhibitors for the Treatment of Fragile X Syndrome and Other Brain Disorders By: Gurney, Mark E.; et al Journal of Medicinal Chemistry (2019), 62(10), 4884-4901.
1.1 Reagents: Triethylamine Solvents: Dimethylformamide; 24 h, 110 °C		

Scheme 42 (1 Reaction)

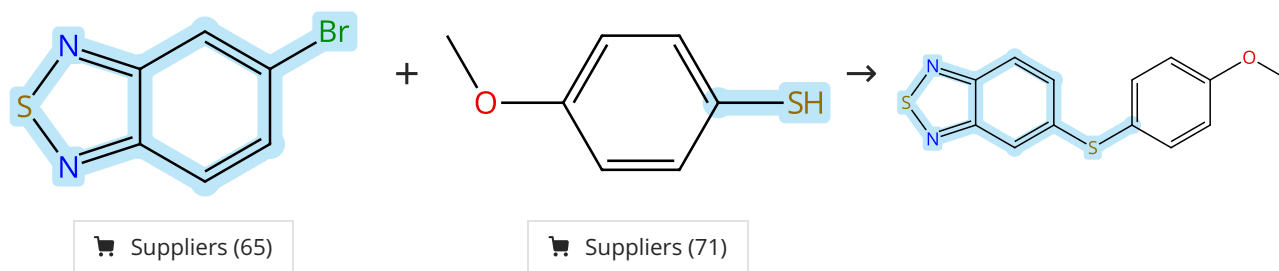
Steps: 1 Yield: 100%



31-614-CAS-40340397	Steps: 1 Yield: 100%	Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling By: Carraro, Massimo; et al Molecules (2024), 29(8), 1714.
1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C		
Experimental Protocols		

Scheme 43 (1 Reaction)

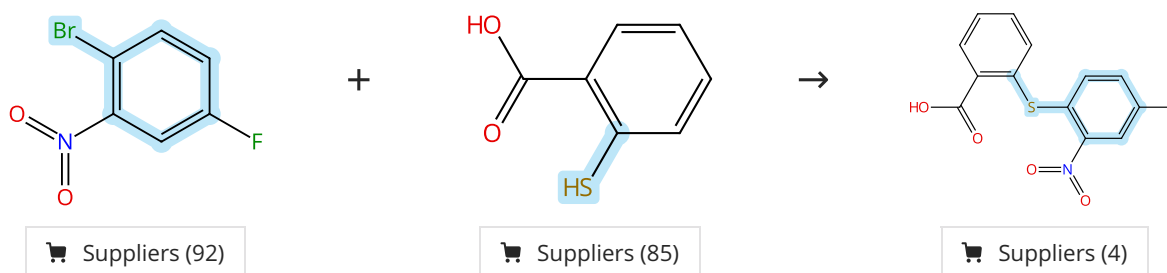
Steps: 1 Yield: 100%



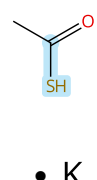
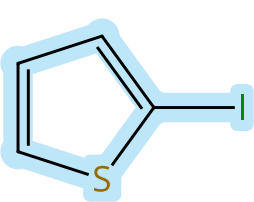
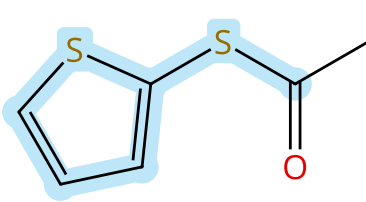
31-171-CAS-21928127	Steps: 1 Yield: 100%	Efficient palladium-catalyzed C-S cross-coupling reaction of benzo-2,1,3-thiadiazole at C-5-position: A potential class of A ChE inhibitors By: dos Santos, Beatriz F.; et al Applied Organometallic Chemistry (2020), 34(7), e5650.
1.1 Reagents: Potassium carbonate Catalysts: Bis(L-prolinato-κN,κO)palladium Solvents: Dimethylformamide; 6 h, 100 °C		
Experimental Protocols		

Scheme 44 (1 Reaction)

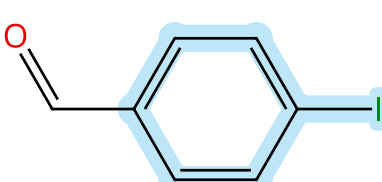
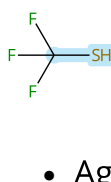
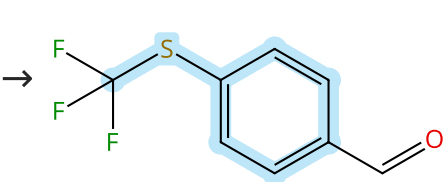
Steps: 1 Yield: 100%



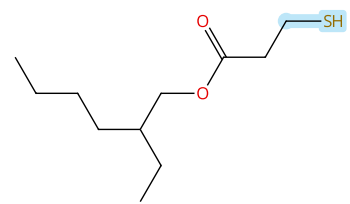
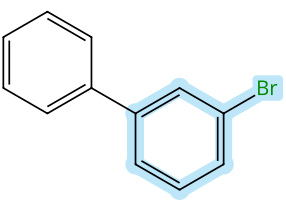
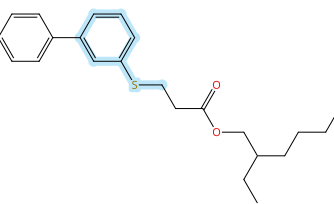
31-171-CAS-20665816 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine, <i>N</i>-Methyl-2-pyrrolidone Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tris (dibenzylideneacetone)dipalladium; 28 h, 85 °C	Preparation of the Serotonin Transporter PET Radiotracer 2-({2-[(Dimethylamino)methyl]phenyl}thio)-5-[¹⁸F]fluoroaniline (4-[¹⁸F]ADAM): Probing Synthetic and Radiosynthetic Methods By: Milicevic Sephton, Selena; et al Synthesis (2019), 51(23), 4374-4384.
--	---

Scheme 45 (1 Reaction) Steps: 1 Yield: 100%		
 • K  Suppliers (84)	 • K  Suppliers (79)	  Suppliers (5)

31-614-CAS-40340410 Steps: 1 Yield: 100% 1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C Experimental Protocols	Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling By: Carraro, Massimo; et al Molecules (2024), 29(8), 1714.
--	--

Scheme 46 (1 Reaction) Steps: 1 Yield: 100%		
 • Ag(I)  Suppliers (93)	 • Ag(I)  Suppliers (51)	  Suppliers (69)

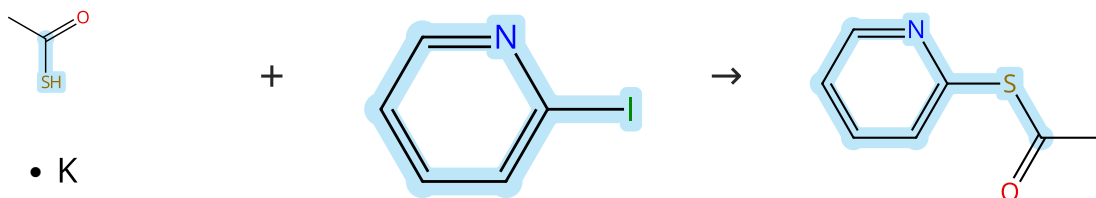
31-614-CAS-31383911 Steps: 1 Yield: 100% 1.1 Catalysts: [2-[Bis(tricyclo[3.3.1.1^{3,7}]dec-1-yl)phosphino-κP]-<i>N</i>, <i>N</i>-dimethylbenzenamine]chlorogold Solvents: 1,2-Dichloroethane; 1 min, rt 1.2 Reagents: Silver hexafluoroantimonate; 1 min, rt; 1.5 h, rt → 70 °C Experimental Protocols	Gold (I/III)-Catalyzed Trifluoromethylthiolation and Trifluoromethylselenenation of Organohalides By: Mudshinge, Sagar R.; et al Angewandte Chemie, International Edition (2022), 61(12), e202115687.
--	---

Scheme 47 (1 Reaction) Steps: 1 Yield: 100%		
  Suppliers (58)	  Suppliers (77)	  Supplier (1)

31-171-CAS-21642227 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: Toluene; 18 h, 110 °C	2-Aminomethylene-5-sulfonylthiazole Inhibitors of Lysyl Oxidase (LOX) and LOXL2 Show Significant Efficacy in Delaying Tumor Growth By: Smithen, Deborah A.; et al Journal of Medicinal Chemistry (2020), 63(5), 2308-2324.
---	---

Scheme 48 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

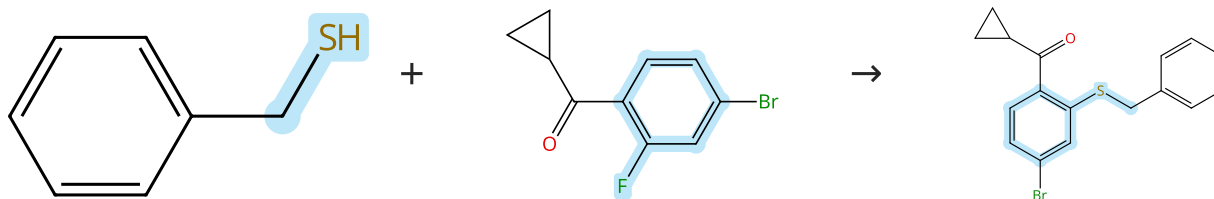
Suppliers (80)

Suppliers (5)

31-614-CAS-40340404 Steps: 1 Yield: 100% 1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C Experimental Protocols	Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling By: Carraro, Massimo; et al Molecules (2024), 29(8), 1714.
--	---

Scheme 49 (1 Reaction)

Steps: 1 Yield: 100%



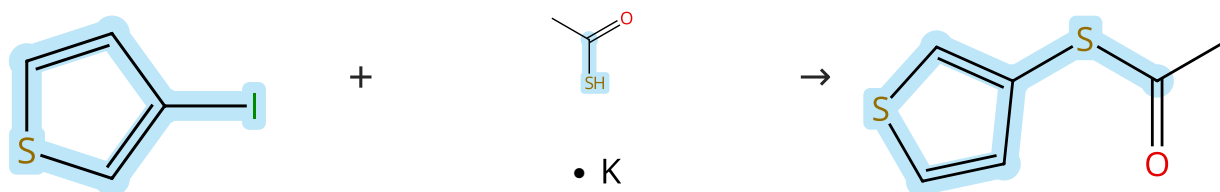
Suppliers (53)

Suppliers (52)

31-171-CAS-19716778 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Tetrahydrofuran; rt; 5 min, rt 1.2 Solvents: Tetrahydrofuran; 30 min, rt Experimental Protocols	Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1<i>H</i>-Pyrazolo[4,3-<i>b</i>]pyridin-3-yl)amino)-benzo[<i>d</i>]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄) By: Bollinger, Sean R.; et al Journal of Medicinal Chemistry (2019), 62(1), 342-358.
--	--

Scheme 50 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

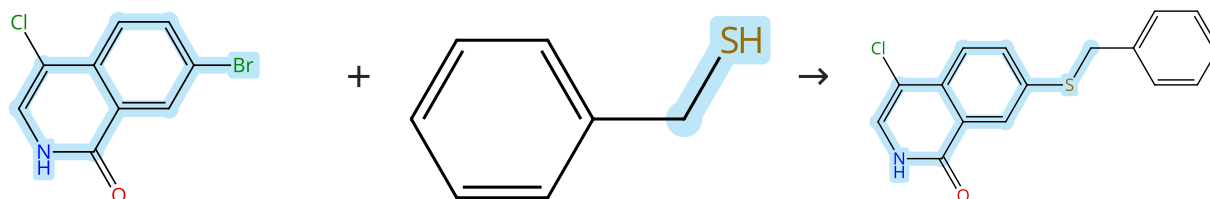
Suppliers (84)

Suppliers (7)

31-614-CAS-40340408	Steps: 1 Yield: 100%	Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling
1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C		By: Carraro, Massimo; et al
Experimental Protocols		Molecules (2024), 29(8), 1714.

Scheme 51 (1 Reaction)

Steps: 1 Yield: 100%



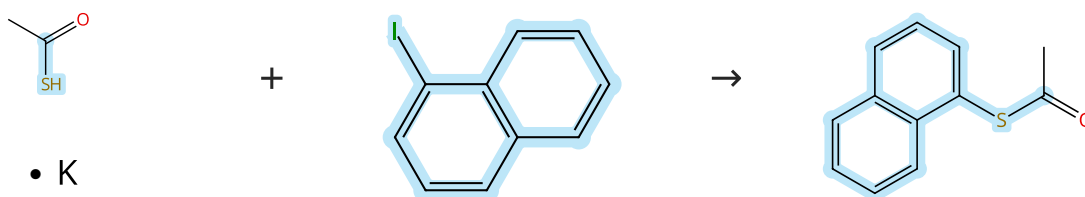
Suppliers (29)

Suppliers (53)

31-171-CAS-19828755	Steps: 1 Yield: 100%	Cell-Active Small Molecule Inhibitors of the DNA-Damage Repair Enzyme Poly(ADP-ribose) Glycohydrolase (PARG): Discovery and Optimization of Orally Bioavailable Quinazolinone Sulfonamides
1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; 20 min, 80 °C		By: Waszkowycz, Bohdan; et al
Experimental Protocols		Journal of Medicinal Chemistry (2018), 61(23), 10767-10792.

Scheme 52 (1 Reaction)

Steps: 1 Yield: 100%



• K

Suppliers (84)

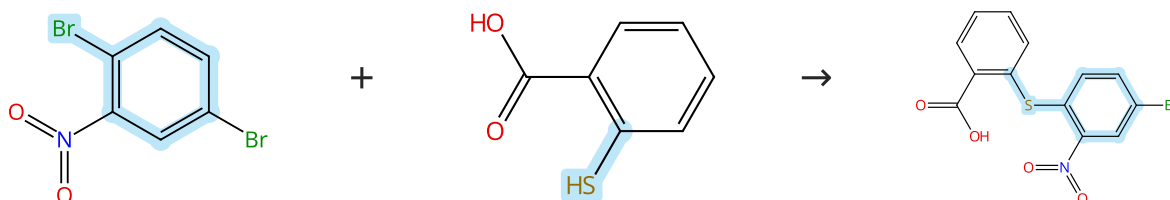
Suppliers (86)

Suppliers (4)

31-614-CAS-40340396	Steps: 1 Yield: 100%	Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling
1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C		By: Carraro, Massimo; et al
Experimental Protocols		Molecules (2024), 29(8), 1714.

Scheme 53 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

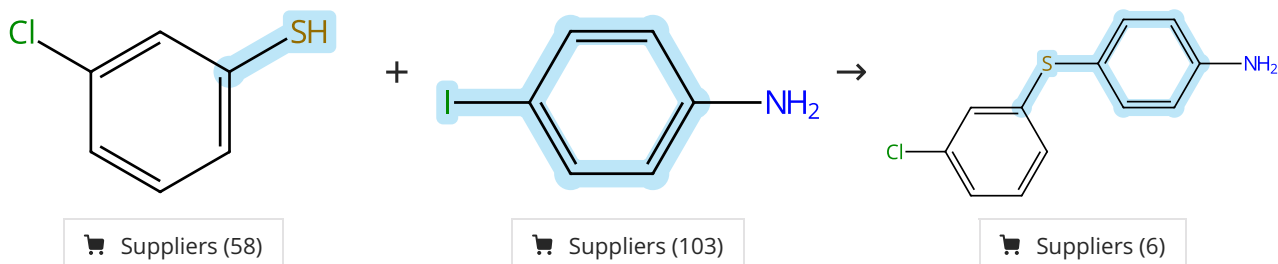
Suppliers (85)

Suppliers (3)

31-171-CAS-20665818 Steps: 1 Yield: 100% 1.1 Reagents: Triethylamine, <i>N</i>-Methyl-2-pyrrolidone Catalysts: 1,1-Bis(diphenylphosphino)ferrocene, Tris (dibenzylideneacetone)dipalladium; 21 h, 85 °C	Preparation of the Serotonin Transporter PET Radiotracer 2-({2-[(Dimethylamino)methyl]phenyl}thio)-5-[¹⁸F]fluoroaniline (4-[¹⁸F]ADAM): Probing Synthetic and Radiosynthetic Methods By: Milicevic Sephton, Selenia; et al Synthesis (2019), 51(23), 4374-4384.
---	--

Scheme 54 (1 Reaction)

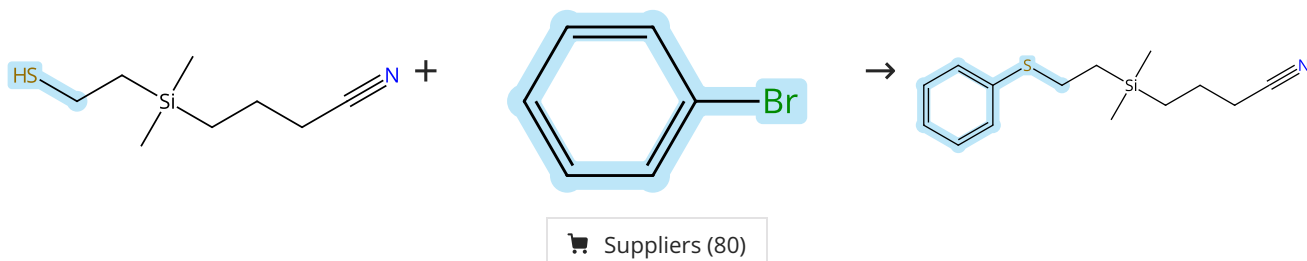
Steps: 1 Yield: 100%



31-614-CAS-31632128 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Copper oxide (CuO) Solvents: Toluene; 24 h, 110 °C Experimental Protocols	Rationale, synthesis and biological evaluation of substituted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site By: Gagne-Boulet, Mathieu; et al Journal of Molecular Structure (2022), 1259, 132691.
---	---

Scheme 55 (1 Reaction)

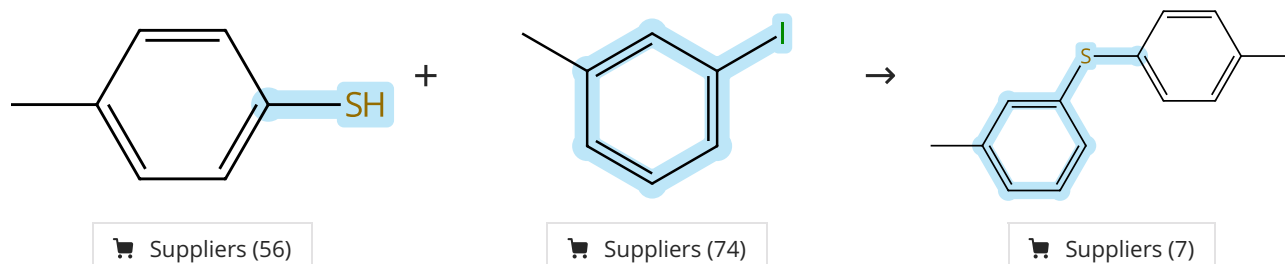
Steps: 1 Yield: 100%



31-171-CAS-20938535 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: 1,4-Dioxane; 5 min, rt 1.2 Catalysts: Dichlorobis(triphenylphosphine)palladium, 1,1'-(9,9-Dimethyl-9<i>H</i>-xanthene-4,5-diyl)bis[1,1-diphenylphosphine]; 12 h, 110 °C	2-(3-Cyanopropyl)dimethylsilyl)ethyl as a Polar Sulfur Protecting Group By: Bannwart, Linda M.; et al Synthesis (2019), 51(22), 4153-4164.
--	--

Scheme 56 (1 Reaction)

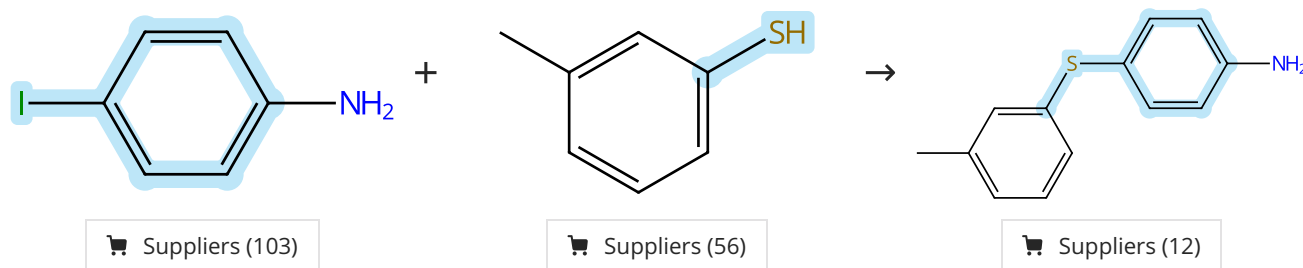
Steps: 1 Yield: 100%



31-171-CAS-23873550	Steps: 1 Yield: 100%	Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides
1.1 Reagents: Potassium hydroxide Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C		By: Zhou, Wen-Yan; et al
1.2 Catalysts: Cuprous iodide; 12 h, 110 °C		Journal of Chemical Crystallography (2021), 51(3), 301-310.
Experimental Protocols		

Scheme 57 (1 Reaction)

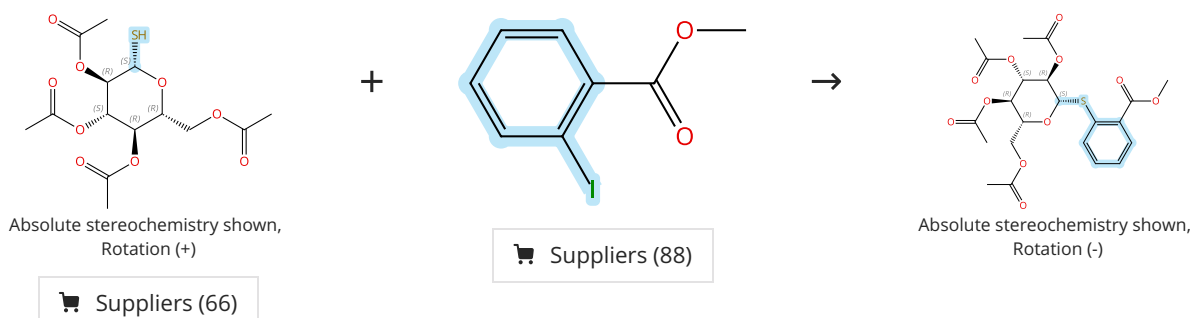
Steps: 1 Yield: 100%



31-614-CAS-31632126	Steps: 1 Yield: 100%	Rationale, synthesis and biological evaluation of substituted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site
1.1 Reagents: Cesium carbonate Catalysts: Copper oxide (CuO) Solvents: Toluene; 24 h, 110 °C		By: Gagne-Boulet, Mathieu; et al
Experimental Protocols		Journal of Molecular Structure (2022), 1259, 132691.

Scheme 58 (1 Reaction)

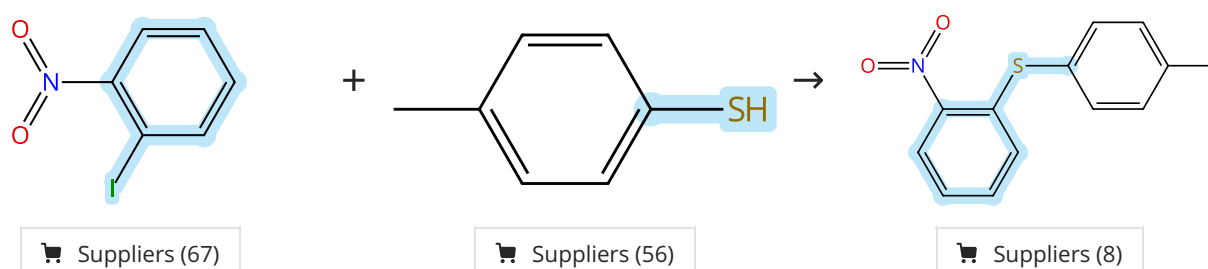
Steps: 1 Yield: 100%



31-171-CAS-13577769	Steps: 1 Yield: 100%	Diversity-oriented synthesis of fused thioglycosyl benzo[e][1,4]oxathiepin-5-ones and benzo[f][1,4]thiazepin-5(2H)-ones by a sequence of palladium-catalyzed glycosyl thiol arylation and deprotection-lactonization reactions
1.1 Reagents: Triethylamine Catalysts: Palladium diacetate, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; 1 h, 100 °C		By: AL-Shuaeeb, Riyadh Ahmed Atto; et al
Experimental Protocols		Organic & Biomolecular Chemistry (2015), 13(44), 10904-10916.

Scheme 59 (1 Reaction)

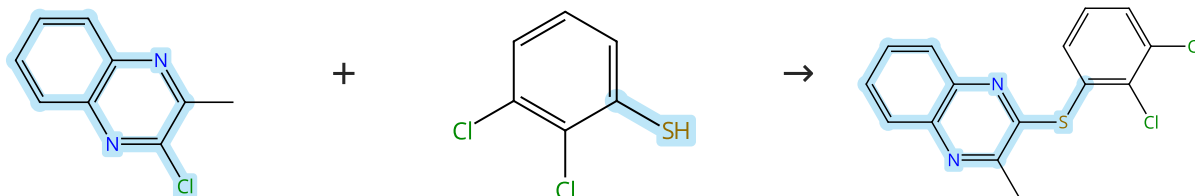
Steps: 1 Yield: 100%



31-171-CAS-18817230 Steps: 1 Yield: 100% 1.1 Reagents: Potassium carbonate Catalysts: Bis(L-prolinato-κN,κO)palladium Solvents: Ethanol; 6 h, 70 °C	C-S Cross-Coupling Reaction Using a Recyclable Palladium Prolinate Catalyst under Mild and Green Conditions By: Santos, Beatriz F.; et al ChemistrySelect (2017), 2(28), 9063-9068.
--	---

Scheme 60 (1 Reaction)

Steps: 1 Yield: 100%



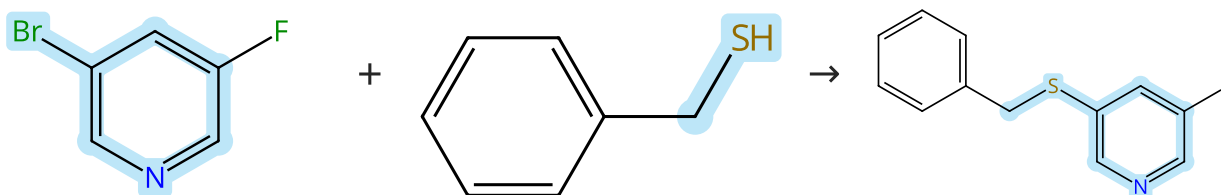
Suppliers (68)

Suppliers (69)

31-614-CAS-31987811 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Solvents: Dimethylformamide; overnight, 70 °C Experimental Protocols	Antiplasmodial 2-thiophenoxy-3-trichloromethyl quinoxalines target the apicoplast of Plasmodium falciparum By: Amrane, Dyhia; et al European Journal of Medicinal Chemistry (2021), 224, 113722.
--	--

Scheme 61 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (86)

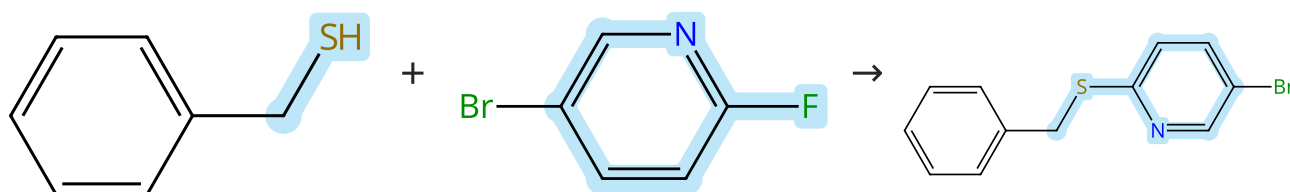
Suppliers (53)

Suppliers (10)

31-171-CAS-17516766 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: Toluene; 3 h, 80 °C Experimental Protocols	Exploration of pyrrole derivatives to find an effective potassium-competitive acid blocker with moderately long-lasting suppression of gastric acid secretion By: Nishida, Haruyuki; et al Bioorganic & Medicinal Chemistry (2017), 25(13), 3447-3460.
--	--

Scheme 62 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (53)

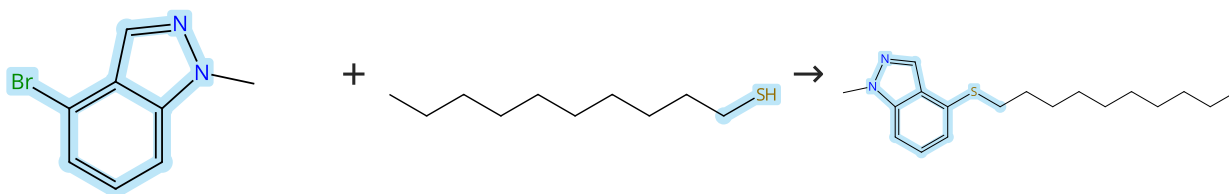
Suppliers (103)

Suppliers (24)

31-171-CAS-4359940	Steps: 1 Yield: 100%	Boronic species as promising inhibitors of the <i>Staphylococcus aureus</i> NorA efflux pump: study of 6- substituted pyridine-3- boronic acid derivatives
1.1 Reagents: Sodium hydride Solvents: Tetrahydrofuran; rt; 30 min, rt; 1 h, rt → reflux		By: Fontaine, Fanny; et al
1.2 12 h, reflux		European Journal of Medicinal Chemistry (2015), 95, 185-198.
Experimental Protocols		

Scheme 63 (1 Reaction)

Steps: 1 Yield: 100%



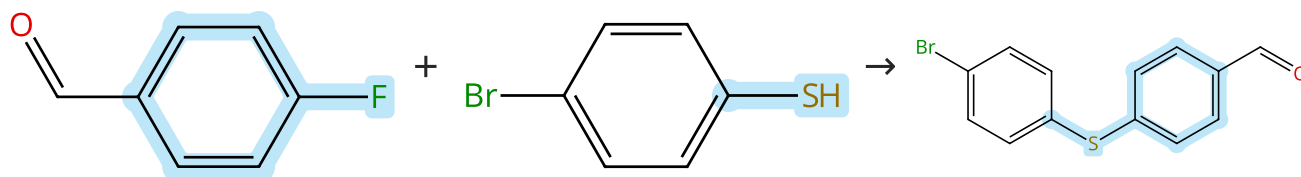
Suppliers (68)

Suppliers (56)

31-171-CAS-20461518	Steps: 1 Yield: 100%	Monophosphine Ligands Promote Pd-Catalyzed C-S Cross-Coupling Reactions at Room Temperature with Soluble Bases
1.1 Reagents: Triethylamine Catalysts: Palladium, bromo[[3,6-dimethoxy-2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl-κC ¹ ']bis(1,1-dimethylethyl)phosphine-κP][4-[[2-(trimethylsilyl)ethoxy]carbonyl]phenyl]-, (SP-4-2)- Solvents: Tetrahydrofuran; 2 h, rt		By: Xu, Jessica; et al ACS Catalysis (2019), 9(7), 6461-6466.

Scheme 64 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

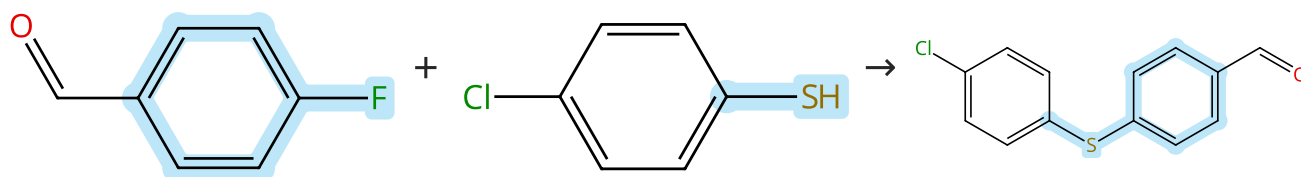
Suppliers (77)

Suppliers (7)

31-614-CAS-41211202	Steps: 1 Yield: 100%	Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C		By: Hasdemir, Belma; et al
Experimental Protocols		ChemistrySelect (2024), 9(26), e202401028.

Scheme 65 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

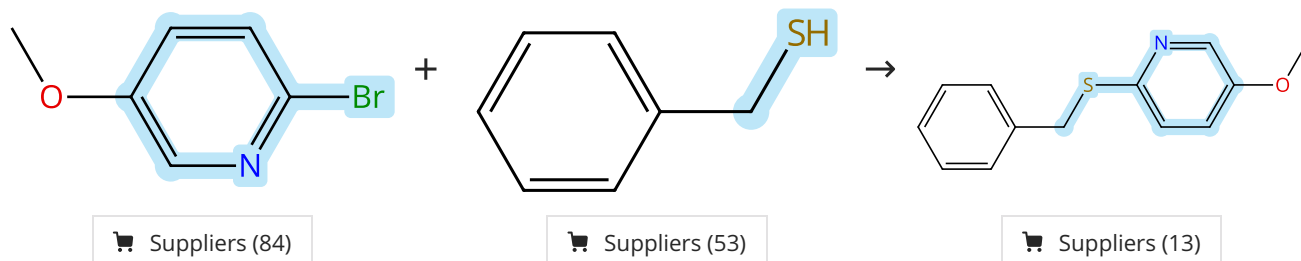
Suppliers (64)

Suppliers (21)

31-614-CAS-41211205	Steps: 1 Yield: 100%	Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C		By: Hasdemir, Belma; et al
Experimental Protocols		ChemistrySelect (2024), 9(26), e202401028.

Scheme 66 (1 Reaction)

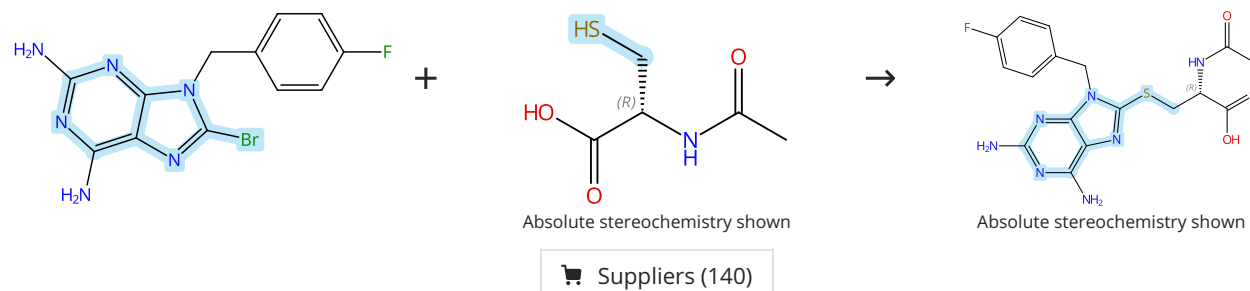
Steps: 1 Yield: 100%



31-171-CAS-17516752	Steps: 1 Yield: 100%	Exploration of pyrrole derivatives to find an effective potassium-competitive acid blocker with moderately long-lasting suppression of gastric acid secretion
1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: Toluene; 26 h, 80 °C		By: Nishida, Haruyuki; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2017), 25(13), 3447-3460.

Scheme 67 (1 Reaction)

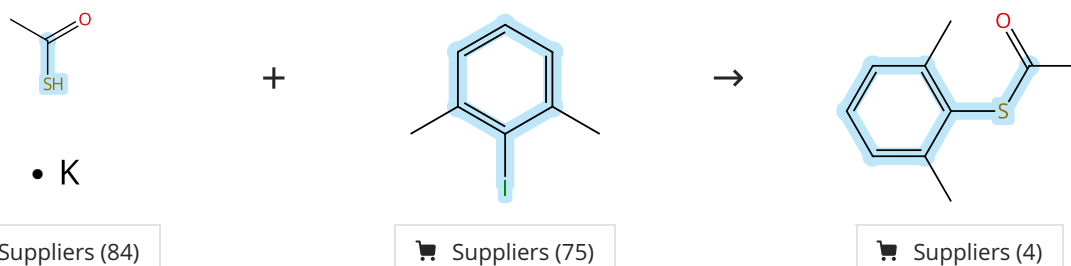
Steps: 1 Yield: 100%



31-614-CAS-37526712	Steps: 1 Yield: 100%	Second-Generation AUTACs for Targeted Autophagic Degradation
1.1 Reagents: Potassium carbonate Solvents: Dimethyl sulfoxide; rt; 2 h, 80 °C		By: Takahashi, Daiki; et al
1.2 Reagents: Hydrochloric acid Solvents: Water; pH 4		Journal of Medicinal Chemistry (2023), 66(17), 12342-12372.
Experimental Protocols		

Scheme 68 (1 Reaction)

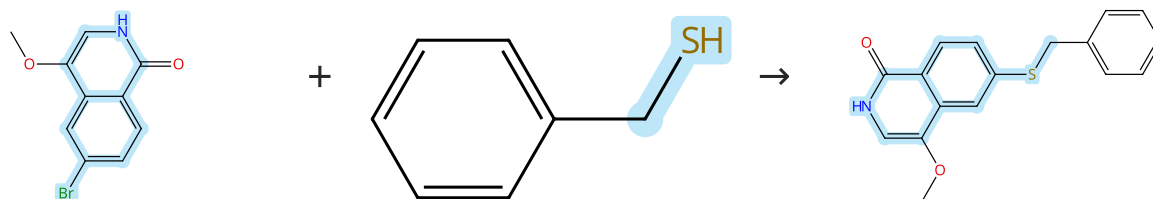
Steps: 1 Yield: 100%



31-614-CAS-40340393	Steps: 1 Yield: 100%	Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling
1.1 Catalysts: 1,10-Phenanthroline, Cuprous iodide Solvents: Cyclopentyl methyl ether; rt → 100 °C; 24 h, 100 °C		By: Carraro, Massimo; et al
Experimental Protocols		Molecules (2024), 29(8), 1714.

Scheme 69 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (29)

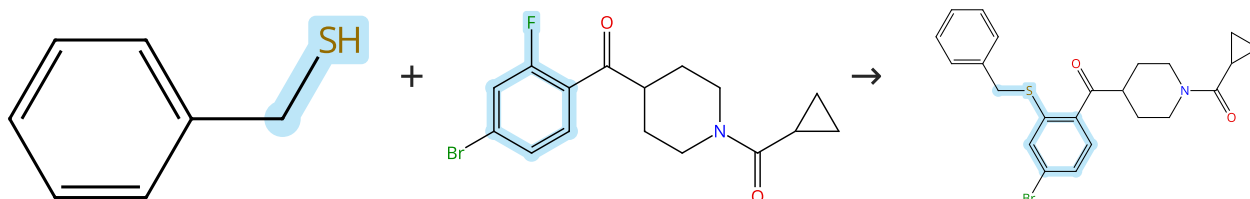
Suppliers (53)

Suppliers (3)

31-171-CAS-17394831	Steps: 1 Yield: 100%	Sulfonamides as Selective Nav1.7 Inhibitors: Optimizing Potency and Pharmacokinetics While Mitigating Metabolic Liabilities
1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; rt → 110 °C; 30 min, 110 °C; 110 °C → rt		By: Weiss, Matthew M.; et al
1.2 Solvents: Water; 10 min, rt		Journal of Medicinal Chemistry (2017), 60(14), 5969-5989.
Experimental Protocols		

Scheme 70 (1 Reaction)

Steps: 1 Yield: 100%

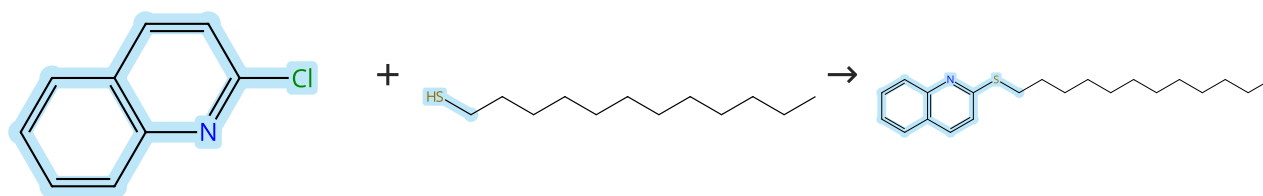


Suppliers (53)

31-171-CAS-19716797	Steps: 1 Yield: 100%	Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄)
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Tetrahydrofuran; rt; 5 min, rt		By: Bollinger, Sean R.; et al
1.2 Solvents: Tetrahydrofuran; 30 min, rt		Journal of Medicinal Chemistry (2019), 62(1), 342-358.
Experimental Protocols		

Scheme 71 (1 Reaction)

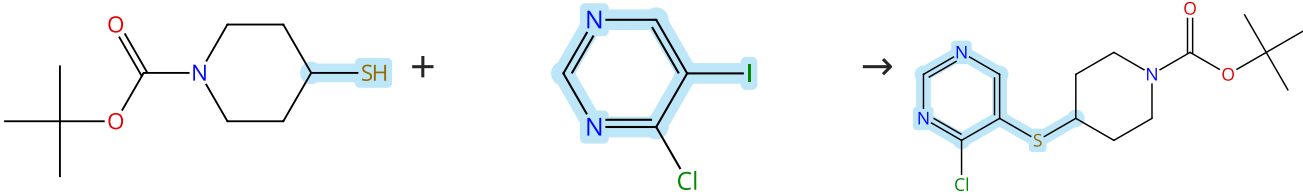
Steps: 1 Yield: 100%



Suppliers (90)

Suppliers (57)

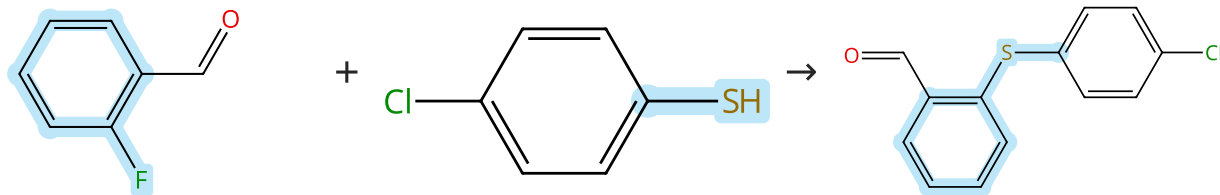
31-614-CAS-37149594	Steps: 1 Yield: 100%	Syntheses of aryl thioethers via aromatic substitution of aryl halides at 0 to 25°C
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Dimethylformamide, Tetrahydrofuran; 5 min, 0 °C; 2 h, 25 °C		By: Tsuzaki, Marina; et al
1.2 Reagents: Ammonium chloride Solvents: Water; 25 °C		Chemical & Pharmaceutical Bulletin (2023), 71(7), 620-623.
Experimental Protocols		

Scheme 72 (1 Reaction)	Steps: 1 Yield: 100%
	
 Suppliers (61)	 Suppliers (70)

31-614-CAS-41112174	Steps: 1 Yield: 100%	Identification of sulfonylpyrimidines as novel selective aldosterone synthase (CYP11B2) inhibitors
1.1 Reagents: Diisopropylethylamine Catalysts: Tris(dibenzylideneacetone)dipalladium, 1,1'-(9,9-Dimethyl-9H-xanthene-4,5-diyl)bis[1,1-diphenylphosphine] Solvents: 1,4-Dioxane; 90 °C		By: Meguro, Masaki; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry (2024), 108, 117775.

Scheme 73 (2 Reactions)

Steps: 1 Yield: 100%



The reaction scheme shows the synthesis of 2-(4-chlorophenylthio)benzaldehyde. The first reactant is 2-fluorobenzaldehyde, a benzene ring with an aldehyde group (-CHO) at position 1 and a fluorine atom (-F) at position 2. The second reactant is 4-chlorobenzenethiol, a benzene ring with a thiol group (-SH) at position 1 and a chlorine atom (-Cl) at position 4. The product is 2-(4-chlorophenylthio)benzaldehyde, which consists of two benzene rings connected by a thioether bridge (-S-). The first benzene ring has an aldehyde group (-CHO) at position 1 and is connected to the sulfur at position 2. The second benzene ring is connected to the sulfur at position 1 and has a chlorine atom (-Cl) at position 4. The reaction is indicated by a plus sign between the reactants and a right-pointing arrow leading to the product.

 Suppliers (83)

 Suppliers (64)

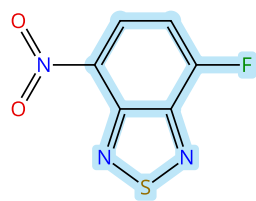
 Suppliers (56)

31-614-CAS-38947796	Steps: 1 Yield: 100%	Intramolecular Friedel-Crafts Reaction with Trifluoroacetic Acid: Synthesizing Some New Functionalized 9-Aryl/Alkyl Thioxanthenes
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C		By: Deniz, Hakan; et al
Experimental Protocols		ACS Omega (2024), 9(11), 12596-12601.

31-171-CAS-19066778	Steps: 1 Yield: 100%	Synthesis of new thioxanthenes by organocatalytic intramolecular Friedel-Crafts reaction
1.1 Reagents: Potassium carbonate Solvents: Dimethylformamide; 2 h, 180 °C		By: Yildiz, Tulay
Experimental Protocols		Synthetic Communications (2018), 48(17), 2177-2188.

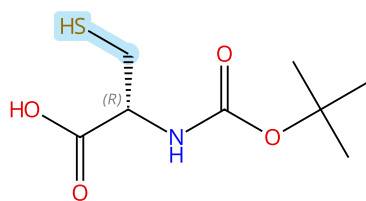
Scheme 74 (1 Reaction)

Steps: 1 Yield: 100%



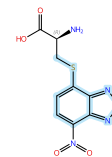
Suppliers (12)

+

Absolute stereochemistry shown,
Rotation (+)

Suppliers (77)

→

• HCl
Absolute stereochemistry shown

31-614-CAS-39061035

Steps: 1 Yield: 100%

Small Fluorogenic Amino Acids for Peptide-Guided Background-Free Imaging

By: de Moliner, Fabio; et al

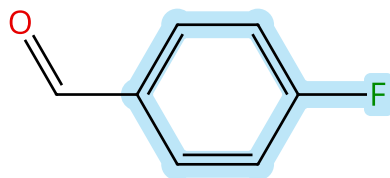
Angewandte Chemie, International Edition (2023), 62(4), e202216231.

- 1.1 Reagents: Sodium bicarbonate
Solvents: Methanol, Water; 55 °C; 20 min, 55 °C
- 1.2 Reagents: Hydrochloric acid
Solvents: 1,4-Dioxane; 30 min, rt

Experimental Protocols

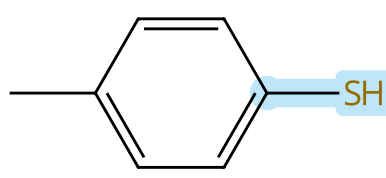
Scheme 75 (1 Reaction)

Steps: 1 Yield: 100%



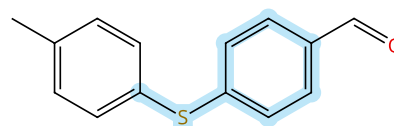
Suppliers (93)

+



Suppliers (56)

→



Suppliers (31)

31-614-CAS-41211203

Steps: 1 Yield: 100%

Synthesis and Characterization of New Bulky α-Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

By: Hasdemir, Belma; et al

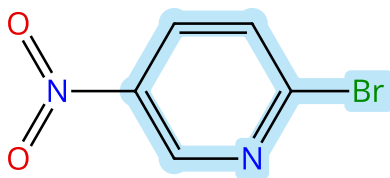
ChemistrySelect (2024), 9(26), e202401028.

- 1.1 Reagents: Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

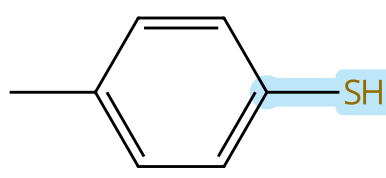
Scheme 76 (1 Reaction)

Steps: 1 Yield: 100%



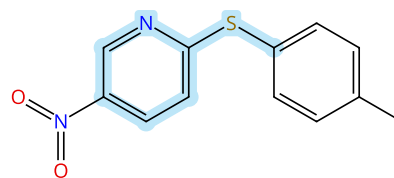
Suppliers (108)

+



Suppliers (56)

→



Suppliers (8)

31-171-CAS-22874141

Steps: 1 Yield: 100%

Synthesis of Pyridylsulfonium Salts and Their Application in the Formation of Functionalized Bipyridines

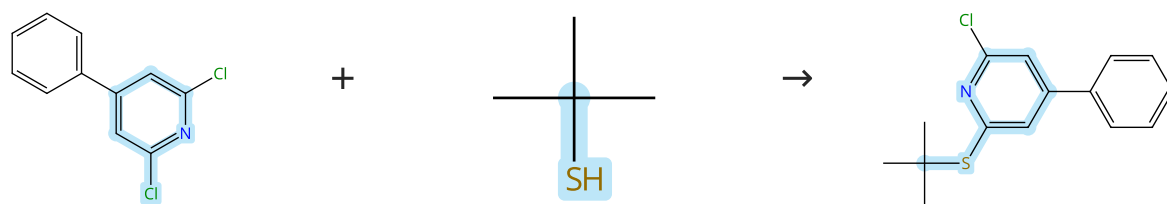
By: Duong, Vincent K.; et al

Organic Letters (2020), 22(21), 8451-8457.

- 1.1 Solvents: Water; 24 h, reflux

Scheme 77 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (47)

Suppliers (50)

Supplier (1)

31-614-CAS-24707749

Steps: 1 Yield: 100%

Crystal Structure and Subsequent Ligand Design of a Nonriboside Partial Agonist Bound to the Adenosine A_{2A} Receptor

By: Amelia, Tasia; et al

Journal of Medicinal Chemistry (2021), 64(7), 3827-3842.

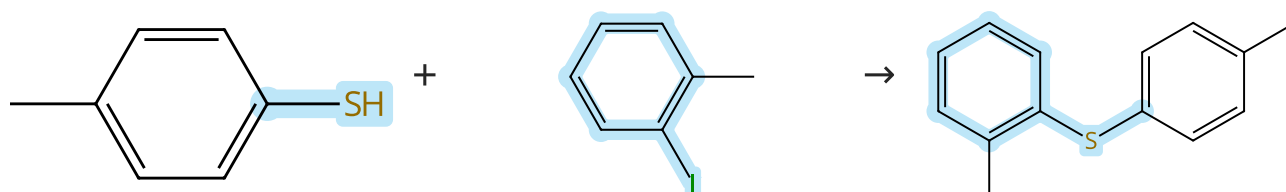
1.1 Reagents: Cesium carbonate

Solvents: Dimethylformamide; overnight, 80 °C

Experimental Protocols

Scheme 78 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (56)

Suppliers (77)

Suppliers (3)

31-171-CAS-23871962

Steps: 1 Yield: 100%

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

Journal of Chemical Crystallography (2021), 51(3), 301-310.

1.1 Reagents: Potassium hydroxide

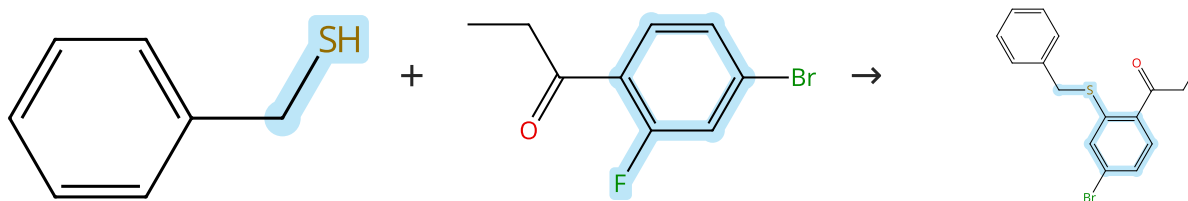
Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C

1.2 Catalysts: Cuprous iodide; 12 h, 110 °C

Experimental Protocols

Scheme 79 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (53)

Suppliers (56)

Supplier (1)

31-171-CAS-19716777

Steps: 1 Yield: 100%

Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄)

By: Bollinger, Sean R.; et al

Journal of Medicinal Chemistry (2019), 62(1), 342-358.

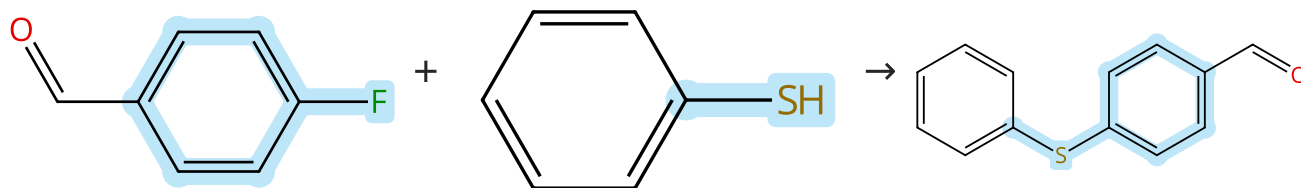
1.1 Reagents: Potassium *tert*-butoxide
Solvents: Tetrahydrofuran; rt; 5 min, rt

1.2 Solvents: Tetrahydrofuran; 30 min, rt

Experimental Protocols

Scheme 80 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (93)

Suppliers (44)

Suppliers (55)

31-614-CAS-41211197

Steps: 1 Yield: 100%

Synthesis and Characterization of New Bulky α -Hydroxy Ketones by N-Heterocyclic Carbene-Catalyzed Intramolecular Benzoin Condensation

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

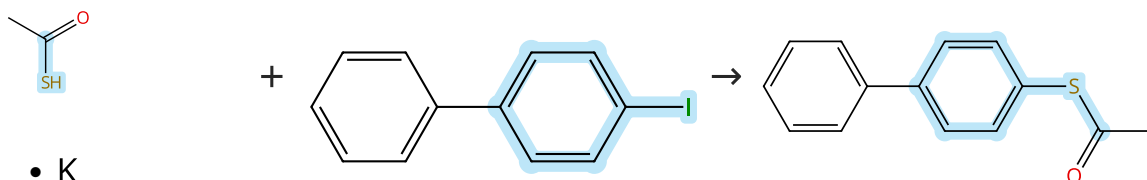
Experimental Protocols

By: Hasdemir, Belma; et al

ChemistrySelect (2024), 9(26), e202401028.

Scheme 81 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (86)

Suppliers (13)

31-614-CAS-40340395

Steps: 1 Yield: 100%

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

1.1 **Catalysts:** 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; rt $\rightarrow$ 100 °C; 24 h, 100 °C

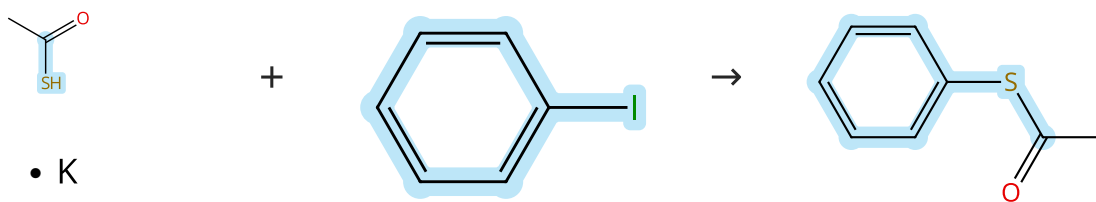
Experimental Protocols

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 82 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (84)

Suppliers (96)

Suppliers (65)

31-614-CAS-40340394

Steps: 1 Yield: 100%

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

1.1 **Catalysts:** 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; 2 h, 100 °C

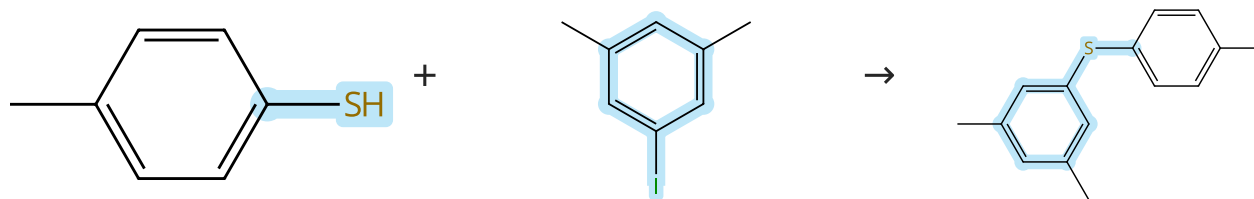
Experimental Protocols

By: Carraro, Massimo; et al

Molecules (2024), 29(8), 1714.

Scheme 83 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (56)

Suppliers (82)

31-171-CAS-23873035

Steps: 1 Yield: 100%

Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides

By: Zhou, Wen-Yan; et al

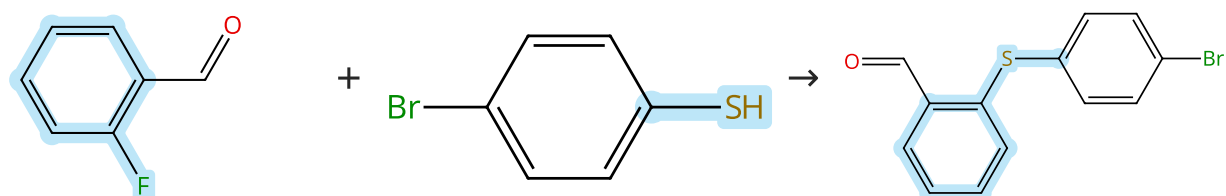
Journal of Chemical Crystallography (2021), 51(3), 301-310.

- 1.1 **Reagents:** Potassium hydroxide
Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C
- 1.2 **Catalysts:** Cuprous iodide; 12 h, 110 °C

Experimental Protocols

Scheme 84 (3 Reactions)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (77)

Suppliers (7)

31-614-CAS-38947803

Steps: 1 Yield: 100%

Intramolecular Friedel-Crafts Reaction with Trifluoroacetic Acid: Synthesizing Some New Functionalized 9-Aryl/Alkyl Thioxanthenes

By: Deniz, Hakan; et al

ACS Omega (2024), 9(11), 12596-12601.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

31-614-CAS-36277482

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

By: Can Uskup, Hacer; et al

ACS Omega (2023), 8(15), 14047-14052.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

31-171-CAS-19066779

Steps: 1 Yield: 100%

Synthesis of new thioxanthenes by organocatalytic intramolecular Friedel-Crafts reaction

By: Yildiz, Tulay

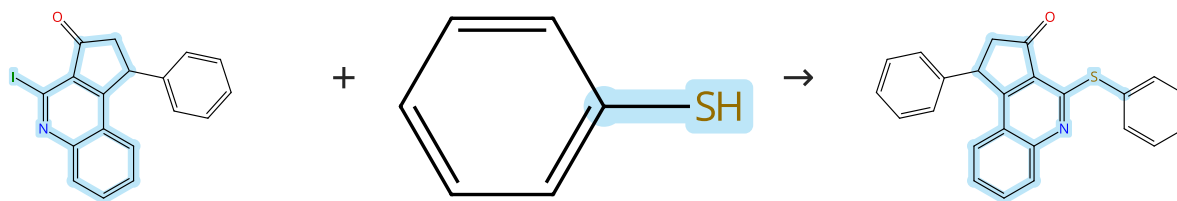
Synthetic Communications (2018), 48(17), 2177-2188.

- 1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

Experimental Protocols

Scheme 85 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (44)

31-171-CAS-16397518

Steps: 1 Yield: 100%

Preparation and Reactions of Indoleninyl Halides: Scaffolds for the Synthesis of Spirocyclic Indole Derivatives

By: Liddon, John T. R.; et al

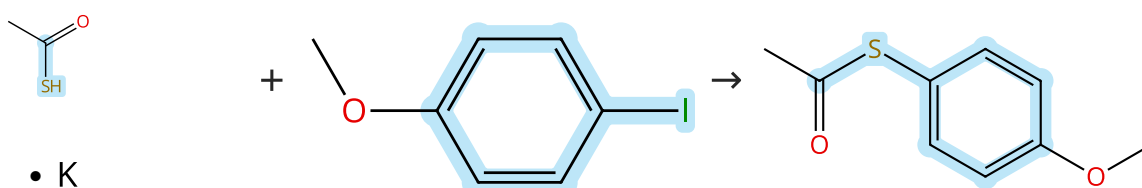
Organic Letters (2016), 18(24), 6328-6331.

1.1 **Reagents:** Cesium carbonate
Solvents: Acetonitrile; 3 h, rt

Experimental Protocols

Scheme 86 (1 Reaction)

Steps: 1 Yield: 100%



• K

Suppliers (84)

Suppliers (89)

Suppliers (18)

31-614-CAS-40340398

Steps: 1 Yield: 100%

Electronic Effects in a Green Protocol for (Hetero) Aryl-S Coupling

By: Carraro, Massimo; et al

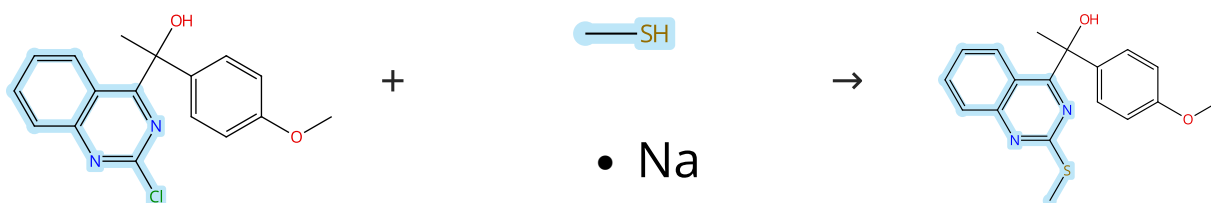
Molecules (2024), 29(8), 1714.

1.1 **Catalysts:** 1,10-Phenanthroline, Cuprous iodide
Solvents: Cyclopentyl methyl ether; 2 h, 100 °C

Experimental Protocols

Scheme 87 (1 Reaction)

Steps: 1 Yield: 100%



• Na

Suppliers (2)

Suppliers (56)

Supplier (1)

31-171-CAS-5569667

Steps: 1 Yield: 100%

Synthesis and Structure-Activity Relationship Study of 1-Phenyl-1-(quinazolin-4-yl)ethanols as Anticancer Agents

By: Kuroiwa, Kenta; et al

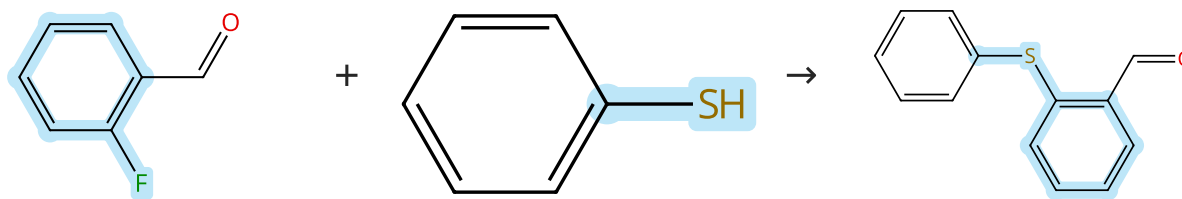
ACS Medicinal Chemistry Letters (2015), 6(3), 287-291.

1.1 **Solvents:** 1,4-Dioxane, Water; 2 h, reflux

Experimental Protocols

Scheme 88 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (44)

Suppliers (43)

31-614-CAS-36277487

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

31-171-CAS-19066775

Steps: 1 Yield: 100%

Synthesis of new thioxanthenes by organocatalytic intramolecular Friedel-Crafts reaction

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

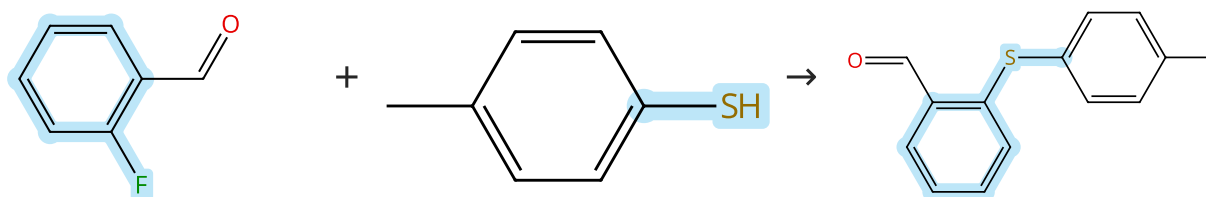
By: Yildiz, Tulay

Experimental Protocols

Synthetic Communications (2018), 48(17), 2177-2188.

Scheme 89 (2 Reactions)

Steps: 1 Yield: 100%



Suppliers (83)

Suppliers (56)

Suppliers (17)

31-614-CAS-36277484

Steps: 1 Yield: 100%

Synthesis of Novel 1,4-Diketone Derivatives and Their Further Cyclization

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

By: Can Uskup, Hacer; et al

Experimental Protocols

ACS Omega (2023), 8(15), 14047-14052.

31-171-CAS-19066777

Steps: 1 Yield: 100%

Synthesis of new thioxanthenes by organocatalytic intramolecular Friedel-Crafts reaction

1.1 **Reagents:** Potassium carbonate
Solvents: Dimethylformamide; 2 h, 180 °C

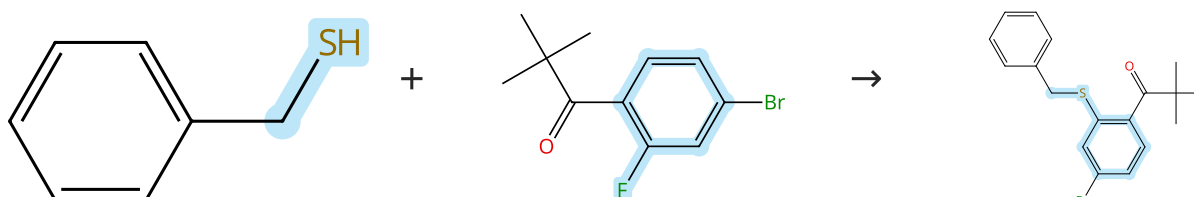
By: Yildiz, Tulay

Experimental Protocols

Synthetic Communications (2018), 48(17), 2177-2188.

Scheme 90 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (53)

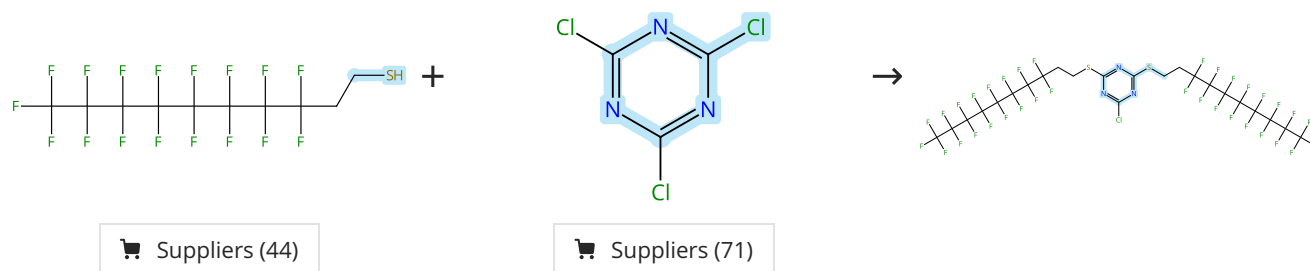
Suppliers (35)

Supplier (1)

31-171-CAS-19716779 Steps: 1 Yield: 100% 1.1 Reagents: Potassium <i>tert</i>-butoxide Solvents: Tetrahydrofuran; rt; 5 min, rt 1.2 Solvents: Tetrahydrofuran; 30 min, rt Experimental Protocols	Discovery, Structure-Activity Relationship, and Biological Characterization of a Novel Series of 6-((1H-Pyrazolo[4,3-b]pyridin-3-yl)amino)-benzo[d]isothiazole-3-carboxamides as Positive Allosteric Modulators of the Metabotropic Glutamate Receptor 4 (mGlu₄) By: Bollinger, Sean R.; et al Journal of Medicinal Chemistry (2019), 62(1), 342-358.
--	---

Scheme 91 (1 Reaction)

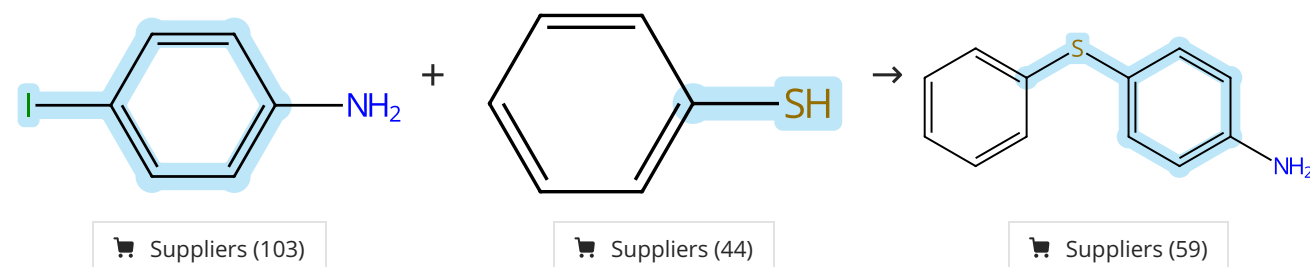
Steps: 1 Yield: 100%



31-171-CAS-19568891 Steps: 1 Yield: 100% 1.1 Reagents: Diisopropylethylamine Solvents: Tetrahydrofuran; 0 °C	Fluoro-tagged osmium and iridium nanoparticles in oxidation reactions By: Santacruz, Lynay; et al Tetrahedron (2018), 74(48), 6890-6895.
---	---

Scheme 92 (1 Reaction)

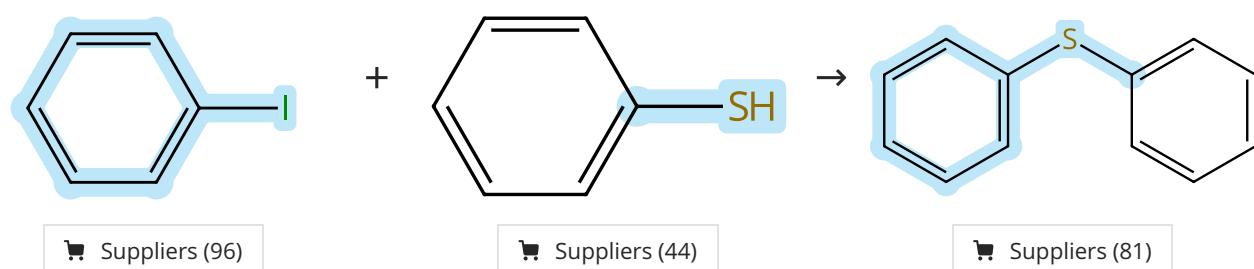
Steps: 1 Yield: 100%



31-614-CAS-31632125 Steps: 1 Yield: 100% 1.1 Reagents: Cesium carbonate Catalysts: Copper oxide (CuO) Solvents: Toluene; 24 h, 110 °C Experimental Protocols	Rationale, synthesis and biological evaluation of substituted 1-(4-(phenylthio)phenyl)imidazolidin-2-one, urea, thiourea and amide analogs and derivatives designed to target the colchicine-binding site By: Gagne-Boulet, Mathieu; et al Journal of Molecular Structure (2022), 1259, 132691.
--	--

Scheme 93 (1 Reaction)

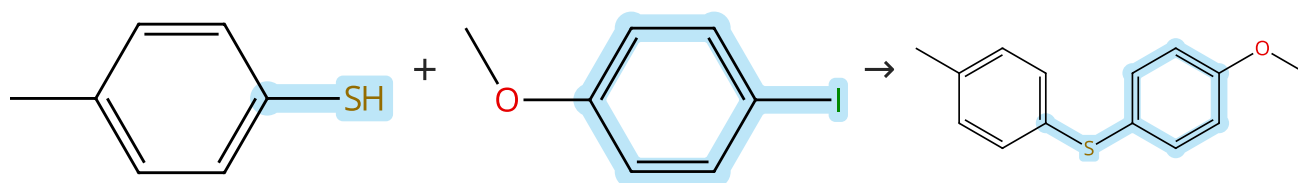
Steps: 1 Yield: 100%



31-614-CAS-42983849	Steps: 1 Yield: 100%	N-Heterocyclic Carbene Switchable Radical- or Benzyne-Mediated Arylation of Thiols Using DMF/KOt-Bu
1.1 Reagents: Potassium <i>tert</i> -butoxide Solvents: Dimethylformamide; 16 h, rt → 60 °C		By: Puschnig, Johannes; et al
Experimental Protocols		Synlett (2025), 36(7), 884-888.

Scheme 94 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (56)

Suppliers (89)

Suppliers (4)

31-171-CAS-23875120	Steps: 1 Yield: 100%	Methods, Syntheses and Characterization of Diaryl, Aryl Benzyl, and Dibenzyl Sulfides
1.1 Reagents: Potassium hydroxide Solvents: Methanol, Polyethylene glycol; 15 min, 70 °C		By: Zhou, Wen-Yan; et al
1.2 Catalysts: Cuprous iodide; 12 h, 110 °C		Journal of Chemical Crystallography (2021), 51(3), 301-310.
Experimental Protocols		